



DOUGLAS COLLEGE
COMMERCE AND BUSINESS
ADMINISTRATION

SafeSight

**Bridging the Analytical Gap through a Predictive Full-Stack
Framework for Proactive Hospitality Safety Management
with reference to**

Fairmont Waterfront



CSIS 4495 002: APPLIED RESEARCH PROJECT

FINAL REPORT

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1. Introduction

Workplace health and safety is a critical domain within Human Resource Management, particularly in high-risk service industries such as hospitality. Hotels operate complex environments where employees are exposed to physical, operational, and ergonomic hazards across departments including Housekeeping, Food and Beverage, Engineering, and Front Office. Effective management of workplace safety is essential to protect employees, ensure regulatory compliance, and maintain operational efficiency (Cooper, 2000).

In British Columbia, organizations are required to comply with occupational health and safety regulations enforced by WorkSafeBC, which mandate accurate incident reporting, investigation, and documentation (WorkSafe BC, 2025). Despite these requirements, many organizations continue to rely on manual or fragmented systems for managing safety data, limiting visibility into incident trends and reducing the ability to proactively address workplace risks.

This research is positioned within the domain of digital workplace safety management systems, combining HR information systems, data visualization, and predictive analytics. SafeSight is a proposed full-stack workplace safety management platform designed specifically for the operational environment of the Fairmont Waterfront. The system aims to support HR and departmental managers by digitizing incident reporting, improving compliance with WorkSafeBC requirements, and providing real-time visibility into workplace safety trends. By combining role-based portals, data visualizations, and predictive analytics, SafeSight enables proactive identification of safety risks and supports informed decision-making within the hotel's hospitality operations.

1.1 Overview of the Background

Organizations across various industries are increasingly adopting digital systems to manage workplace operations, including health and safety management. Traditionally, workplace safety incidents have been recorded using paper-based forms or basic spreadsheet systems, which often result in delayed reporting, inconsistent data, and limited analytical capability. These limitations reduce an organization's ability to identify patterns, assess risks, and take preventive action.

In the hospitality industry, the complexity of operations and the diversity of job roles further increase safety management challenges. Employees frequently work under time pressure, handle

equipment, and perform repetitive physical tasks, making them more vulnerable to workplace injuries. As a result, hotels must manage large volumes of safety-related data while ensuring timely communication between HR teams, managers, and regulatory bodies.

Advancements in web technologies, data visualization, and predictive analytics provide new opportunities to modernize workplace safety management (Stone, et al., 2015)

. Full-stack web applications enable centralized data collection, role-based access through HR and manager portals, and real-time visualization of safety performance. Additionally, predictive analytics allows organizations to move beyond reactive reporting by forecasting potential risks and supporting proactive safety interventions.

This research builds on these technological advancements by proposing a full-stack, role-based workplace safety system that integrates incident management, regulatory reporting, and predictive insights within a single platform.

1.2 Research Context

This research is situated within the growing need for organizations to adopt digital solutions that improve workplace health and safety management while ensuring regulatory compliance. In British Columbia, employers are legally required to follow occupational health and safety standards established by WorkSafeBC, which include timely incident reporting, investigation documentation, and continuous monitoring of workplace risks.

In practice, many organizations particularly in the hospitality industry continue to manage safety incidents using manual processes or fragmented digital tools. These approaches limit data integration, delay reporting, and provide minimal analytical insight for HR professionals and managers. As a result, safety management is often reactive, focusing on incidents after they occur rather than preventing them.

At the same time, advancements in web technologies, data visualization, and predictive analytics have created opportunities to modernize safety management systems. Role-based web applications can support distinct user needs through HR portals and Manager portals, while centralized data storage enables real-time dashboards and automated reporting aligned with regulatory requirements.

Within this context, this research proposes the development of SafeSight, a full-stack, web-based workplace safety platform designed to support incident management, visual analytics, predictive risk assessment, and WorkSafeBC reporting. The system is developed as an applied case study within a hospitality environment, demonstrating how integrated digital solutions can enhance proactive safety management and informed decision-making.

1.3 Problem Statement

In high-pressure luxury hospitality environments like the Fairmont Waterfront, workplace safety management is frequently undermined by a reliance on manual, fragmented, or disconnected systems. These legacy processes create critical "data silos" between frontline staff, department managers, and HR teams, resulting in significant communication latency and a lack of centralized visibility into incident trends. This infrastructure deficit hinders the immediate identification of hazards and delays the production of mandatory WorkSafeBC compliance documentation, often leading to administrative burnout and potential regulatory penalties. Without integrated, role-based digital tools or predictive insights, safety management remains purely reactive, leaving the organization vulnerable to recurring injuries and operational inefficiencies. To address these gaps, SafeSight provides a unified full-stack platform designed to automate reporting, visualize risk trends, and transition the hotel's safety posture to proactive, data-driven prevention.

Research Questions

1. How can a full-stack, role-based safety system improve the speed and accuracy of incident reporting and monitoring?
2. In what ways do interactive dashboards and data visualizations support managerial safety decision-making?
3. To what extent can predictive analytics identify future workplace safety risks based on historical patterns?
4. How can automated reporting modules streamline compliance with WorkSafeBC statutory requirements?

2. Literature Review

Workplace health and safety management has increasingly shifted from manual record-keeping to digital systems that support incident reporting, monitoring, and regulatory compliance (Shah, 2026). Previous studies highlight that effective safety management systems play a critical role in reducing workplace injuries, improving compliance, and supporting proactive risk prevention. With advancements in web technologies, data visualization, and predictive analytics, organizations now have opportunities to enhance traditional safety practices through integrated digital platforms.

Hoteliers normally follow local regulations when designing a hotel's safety and security systems, but which system features hotel internal staff perceive to be the most important and which they expect to find in place remain open questions (Chan & Lam, 2013).

The Evolution of OHSMS in Global and Canadian Contexts

The concept of an Occupational Health and Safety Management System (OHSMS) has matured significantly since its inception in the late 1990s. Early frameworks, such as the OHSAS 18001 and various regional guidelines (British Standard Institution, 1996; ILO, 2001; Gallagher, et al., 2003), established the baseline for managing physical hazards. However, the transition to ISO 45001:2018 marked a paradigm shift, moving safety from a "siloed" administrative task to a core leadership responsibility.

Contextualizing OHSMS for the Canadian Hospitality Sector

For a landmark Canadian institution like the Fairmont Waterfront, the application of OHSMS is no longer just about meeting WorkSafeBC regulations; it is about organizational resilience. In the current post-pandemic landscape (2020–2026), the definition of "occupational health" has expanded to include,

- **Psychological Safety & Mental Wellbeing:** Research indicates that 87% of Canadian hospitality workers have experienced burnout post-pandemic (Future Skills Centre, 2024). Modern OHSMS must now integrate mental health resources to address the high-pressure nature of luxury service.

- **Musculoskeletal Injury (MSI) Prevention:** As high-end hotels return to peak occupancy, the ergonomic risks for housekeeping and guest services specifically regarding repetitive tasks remain a primary focus of Canadian safety audits (WorkSafeBC, 2025).
- **Integration with HRM:** Recent literature suggests that OHSMS is most effective when fused with Human Resource Management (HRM). This "total worker health" approach aligns safety performance with employee retention, which is critical given the persistent labor shortages in the Vancouver hospitality sector (The HUB International, 2026).

2.1 Identification of Research Gaps and the SafeSight Rationale

While the theoretical frameworks for OHSMS are well-established, a critical review of current safety practices in high-end hospitality specifically within the Canadian regulatory context of WorkSafeBC reveals four primary limitations. These gaps form the empirical justification for the development of the SafeSight full-stack system.

The Temporal Gap: Latency in Incident Response

Existing safety management often relies on fragmented communication channels (e.g., paper logs, emails, or verbal reports). In a high-traffic environment like the Fairmont Waterfront, this creates a "latency gap" where hazards remain active between the time of discovery and the time of management notification. Research suggests that "stale data" is a leading contributor to secondary accidents (Nelson, et al., 2013). SafeSight aims to address this by implementing a real-time full-stack architecture that synchronizes field reports with management dashboards instantaneously.

The Integration Gap: Stakeholder Silos

A significant limitation in current hotel operations is the "siloeing" of safety data. HR departments, Department Managers (e.g., Housekeeping, F&B), and Safety Officers often utilize disparate systems to track the same incident. This lack of integration leads to data duplication and inconsistent reporting. SafeSight addresses this by providing a unified "Single Source of Truth," allowing multi-level access to a centralized database that serves the specific needs of each stakeholder from HR compliance to operational hazard mitigation.

The Regulatory Friction Gap: WorkSafeBC Compliance

According to Workers Compensation Act (2026) and WorkSafeBC (2025) Compliance with the Workers Compensation Act requires rigorous documentation and strict filing deadlines (e.g., Form 7 or Preliminary Investigation Reports). The administrative burden of manually transcribing incident data into regulatory formats often leads to "compliance fatigue," where accuracy is sacrificed for speed. SafeSight introduces an automated reporting module designed to map internal incident data directly onto WorkSafeBC-compliant templates, thereby reducing administrative overhead and ensuring 100% filing accuracy.

The Analytical Gap: Descriptive vs. Prescriptive Output

Current OHSMS implementations are largely descriptive they record what happened in the past but offer little insight into how to prevent future occurrences (Lindholm, et al., 2024). There is a lack of localized systems that provide active recommendations. SafeSight seeks to bridge this gap by transitioning from a passive record-keeping tool to a prescriptive analytics platform. By analyzing historical incident trends within specific Fairmont departments, the system generates actionable safety recommendations and heatmaps to guide preventative maintenance and training.

2.2 Research Hypotheses

This research aims to verify the following testable predictions through the development of SafeSight.

- Hypothesis 1 (H1): The implementation of a full-stack, real-time reporting system will reduce the time between incident occurrence and management notification by at least 70% compared to manual/paper-based methods.
- Hypothesis 2 (H2): Automated report generation will significantly decrease the rate of human error and missing data fields in WorkSafeBC documentation.
- Hypothesis 3 (H3): The use of heatmaps and predictive analytics will enable department managers to identify high-risk areas (e.g., the kitchen or loading docks) before recurring incidents occur, leading to a measurable decrease in "Lost Time" accidents.

2.3 Research Assumptions

For this study to be valid, the following conditions are assumed,

- **Stakeholder Participation:** It is assumed that staff and managers at the Fairmont Waterfront will provide honest and timely data when using the SafeSight interface.
- **Data Consistency:** It is assumed that historical incident data provided by the hotel for the predictive module is accurate and representative of typical operations.
- **Infrastructure Reliability:** It is assumed that the hotel's existing internal network can support a web-based, full-stack application without significant downtime.
- **Technical Literacy:** It is assumed that the target users (HR, Managers, and Staff) possess the basic digital literacy required to navigate a role-based dashboard.

2.4 Potential Benefits of SafeSight

The successful development of SafeSight offers significant advantages to the Fairmont Waterfront.

For HR and Senior Management

- **Unified Oversight:** A single dashboard providing a "bird's-eye view" of safety across all departments.
- **Reduced Liability:** Faster, more accurate WorkSafeBC filing reduces the risk of fines and increases the hotel's standing during external audits.

For Department Managers (F&B, Housekeeping, Maintenance)

- **Reduced Admin Burden:** Automating the "paperwork" allows managers to spend more time on guest experience and staff training.
- **Data-Driven Decisions:** Instead of guessing where the risks are, managers can use SafeSight heatmaps to allocate resources or safety signage to specific "hotspots."

For the Fairmont Waterfront Organization

- **Brand Excellence:** Reinforces Fairmont's reputation as a "Top Employer" in Canada by demonstrating a cutting-edge commitment to employee wellbeing.
- **Financial Savings:** By predicting and preventing injuries, the hotel reduces costs associated with worker compensation claims and temporary staffing.

3. Summary of the Research Project

The SafeSight research project was developed as an integrated solution to modernize workplace health and safety management within the Fairmont Waterfront. The project addresses a key industry challenge: the reliance on manual safety logs and reactive reporting systems that limit an organization's ability to prevent incidents before they occur. By combining full-stack web development with artificial intelligence, SafeSight transforms traditional safety practices into a proactive, data-driven decision-making system.

At its core, the platform uses a Random Forest machine learning model to analyze historical incident data and predict high-risk scenarios across different hotel departments. Rather than focusing solely on accuracy, the model was optimized to prioritize recall, ensuring that potential hazards are rarely missed. These predictions are presented through an interactive dashboard that not only highlights risk levels but also provides prescriptive strategic actions, bridging the gap between data analysis and real-world managerial decisions.

In addition to predictive capabilities, SafeSight improves operational efficiency by automating compliance processes such as WorkSafeBC reporting, significantly reducing administrative time. The system also leverages a structured MySQL database to maintain data integrity, enabling advanced features like departmental risk profiling and performance tracking. User experience was a central consideration, with the interface designed to be intuitive for non-technical staff, ensuring high adoption and usability.

Overall, the project demonstrates how AI-driven systems can shift organizations from reactive incident reporting to proactive risk prevention. SafeSight positions the Fairmont Waterfront to foster a culture of safety, efficiency, and continuous improvement, aligning with the broader goal of achieving a "Zero Harm" workplace environment.

3.1 Core Objectives

The primary goal is to transition from reactive to proactive safety management through five key pillars:

- ***System Development***

Creating a role-based (Managers, HR) interface for real-time incident logging.

- ***Visualization***

Using interactive heatmaps and dashboards to identify high-risk hotel zones (e.g., kitchens vs. lobby).

- ***Predictive Modeling***

Utilizing machine learning to forecast potential safety "hotspots" based on historical data.

- ***Compliance Automation***

Reducing administrative burden by at least 50% through automated WorkSafeBC report generation.

- ***Evaluation***

Validating the system's efficiency and user ease-of-use through simulated deployment.

3.2 Technical Framework & Methodology

The project follows an Evolutionary Prototyping Model within an Applied Design Research (ADR) framework. The development is structured into a four-phase System Development Life Cycle (SDLC):

1. **Phase 1 (Analysis):** Gap analysis of current manual logs and mapping WorkSafeBC regulatory requirements.
2. **Phase 2 (Development):** Building the three-tier architecture (Node.js/Express backend, MySQL database, and JavaScript frontend).
3. **Phase 3 (Intelligence):** Integrating a Python/Flask prediction module using scikit-learn to analyze 478 industry-specific injury records.
4. **Phase 4 (Validation):** Quantitative benchmarking of reporting speed and Qualitative User Acceptance Testing (UAT).

3.3 Technology Stack

SafeSight leverages a modern, scalable stack to ensure accessibility across hotel departments:

- **Frontend:** HTML, CSS, JavaScript, and **Chart.js** for data visualization.
- **Backend:** **Node.js** and **Express.js** managing REST APIs.

- **Database: MySQL** for relational data storage of incident logs and corrective actions.
- **AI/Analytics: Python (Flask)** and **scikit-learn** for regression analysis and risk forecasting.

4. Expected Impact

The project aims to deliver a robust technological artifact that not only streamlines WorkSafeBC compliance but also provides "Safety Intelligence." By visualizing trends and predicting risks, SafeSight enables Fairmont leadership to make data-driven decisions, ultimately reducing workplace injuries in high-frequency departments like Housekeeping and Food & Beverage.

4. Changes to the proposal

4.1 Justification for Changes

1. Shift to React.js for Frontend Development

- *The Change:* Moving from vanilla JavaScript to React.js for the presentation layer.
- *Justification:* While vanilla JS is functional, the Managerial Dashboards and Heatmaps require complex state management (e.g., filtering incidents by department or date without reloading the page).
- *Timeline:* Phase 1
- *Research & Decision:* Comparisons showed that React's virtual DOM provides faster updates for real-time "hotspot" visualizations. This ensures the UI remains responsive even when handling 478+ historical records.

2. Enhanced Predictive Modeling (Random Forest vs. Linear Regression)

- *The Change:* Upgrading the prediction module from simple regression to Random Forest Classification.
- *Justification:* Initial assumptions that incident frequency followed a linear path were proved incorrect by the OSHA 721110 data. Hotel injuries (burns vs. slips) are categorical and non-linear.
- *Timeline:* Phase 2
- *Research & Decision:* Random Forest handles categorical variables (e.g., "Kitchen," "Lobby," "Laundry") more accurately, providing better "Safety Intelligence" for HR personnel.

3. Implementation of a Decoupled API Architecture

- *The Change:* Separating the Node.js business logic from the Python (Flask) analytics engine.
- *Timeline:* Phase 3
- *Justification:* Mixing Python (for AI) and Node.js (for user management) in a single environment creates dependency conflicts.

- *Research & Decision:* By using a RESTful bridge, the system becomes more stable. Node.js handles the high-concurrency reporting, while the Python Flask service is called only when a manager requests a "Predictive Risk Report."

4. Localized Mapping of OSHA Data

- *The Change:* Moving from using raw OSHA data to Geographically Mapped Synthetic Data.
- *Justification:* Raw OSHA data lacks the physical context of the Fairmont Waterfront. To make the Heatmaps (Objective 2) valuable, the data had to be "mapped" to specific hotel coordinates (e.g., Floor 3, East Wing Kitchen).
- *Timeline:* Phase 1
- *Research & Decision:* This allows for a more realistic User Acceptance Testing (UAT) phase, as managers can see "hotspots" in areas they recognize rather than abstract data points.

5. Refined Compliance Timeline

- *The Change:* Extending the Phase 4 (Validation) timeline by two weeks.
- *Justification:* Initial estimates underestimated the complexity of mapping 30+ data fields to the specific WorkSafeBC Form 7 PDF structure.
- *Research & Decision:* To ensure the "50% reduction in admin time" (Objective 4) is met, more time was allocated to "Field Mapping" to ensure the auto-generated PDF is 100% legally compliant.

6. Strategic Redirect: From Generic Reporting to Regulatory Automation

- *The Change:* Transitioning from a general-purpose incident summary system to a Regulatory Compliance Automation tool focused on the WorkSafeBC EIIR.
- *Justification:* The initial approach provided limited operational value, whereas automating a legally required report significantly reduces administrative workload and increases real-world applicability within the Fairmont Waterfront.
- *Timeline:* Phase 3

- *Research & Decision:* Research into WorkSafeBC reporting requirements revealed the need for highly structured and legally compliant data. To ensure accurate submission and maximize efficiency, the system was redesigned to include detailed data fields such as safety controls, root cause categories, and witness identification. Additionally, the export module was upgraded to generate a high-fidelity PDF aligned with EIIR standards, while integrating the “SafeSight Format” to combine regulatory compliance with organizational branding.

5. Project Planning and Timeline

5.1 Project Responsibilities

As the sole researcher and developer for SafeSight, the role was divided into three primary "hats".

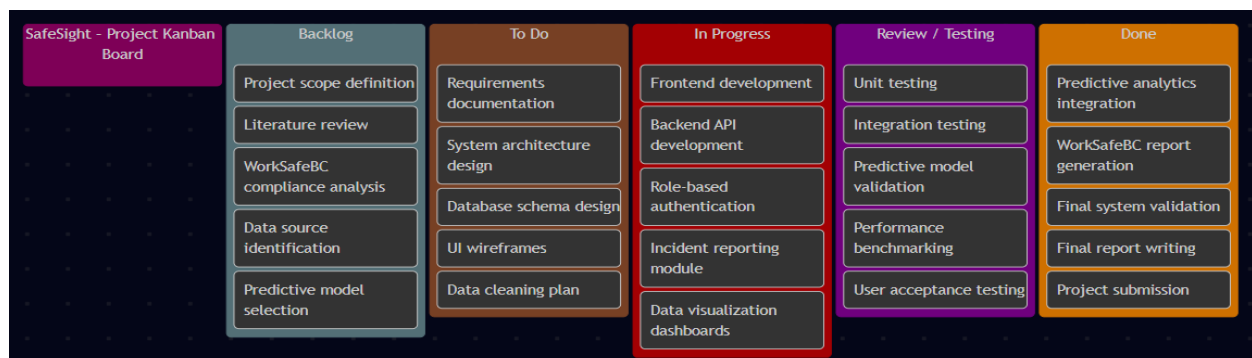
- **System Architect & Full-Stack Developer:** Responsible for the end-to-end technical build, including the React.js frontend, the Node.js API, and the MySQL database design.
- **Data Scientist:** Handle the cleaning of the OSHA 721110 dataset and the development of the Python/Flask predictive module using scikit-learn.
- **Compliance & QA Lead:** Manage the mapping of WorkSafeBC regulatory requirements and conduct all User Acceptance Testing (UAT) to validate the system's efficiency.

Updated Milestones

- **Milestone 1 (Data Foundation):** Successful connection between the React frontend and MySQL database.
- **Milestone 2 (Intelligence Bridge):** Establishing a secure REST API connection between the Node.js server and the Python/Flask prediction engine.
- **Milestone 3 (The "Efficiency" Test):** Successfully generating a WorkSafeBC Form 7 PDF with a single click, populated with real database values.

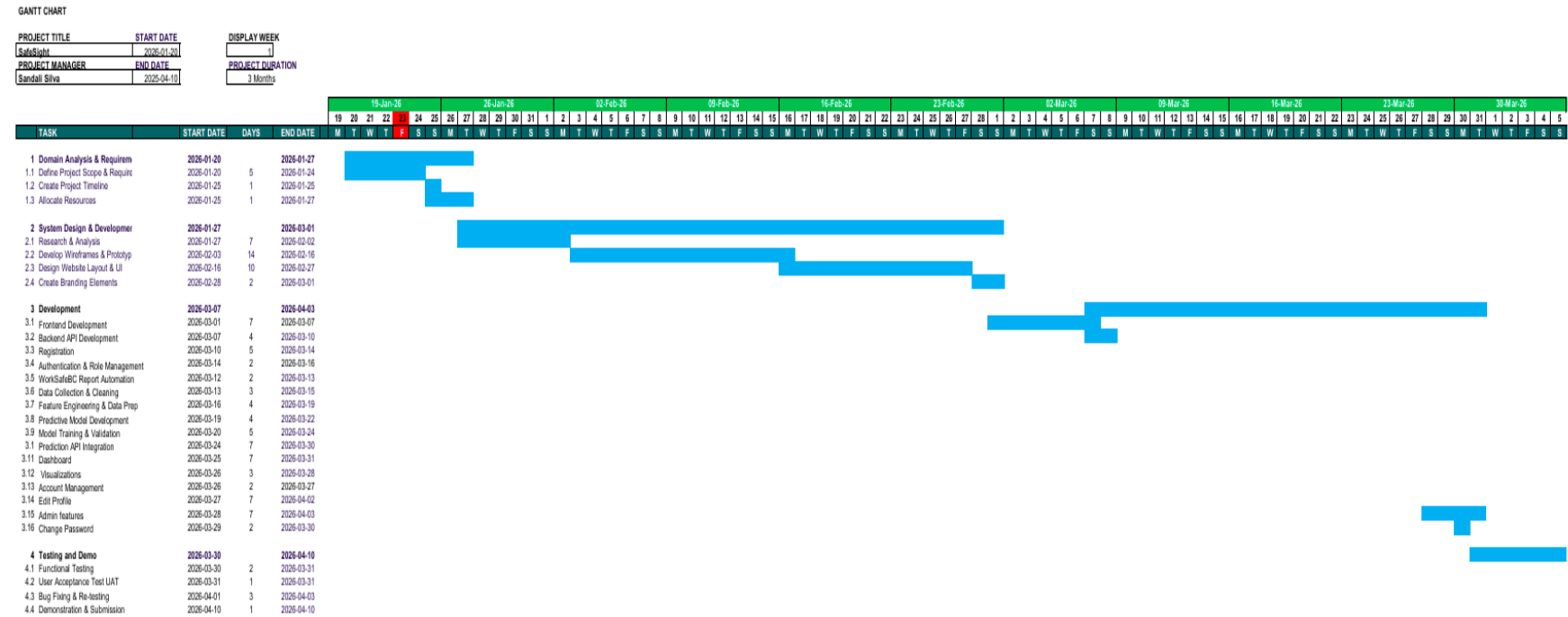
To ensure structured progress and timely completion, this project follows a defined timeline supported by both a Gantt chart and a Kanban board, outlining key phases, milestones, deadlines, and deliverables.

5.1 KANBAN



5.2 Gantt Chart

The following chart visualizes the overlap between the Predictive Modeling and Compliance Mapping, ensuring the backend is ready before the final validation begins.



5.3 Research Project Timeline and Milestones (SafeSight)

Phase	Duration (Weeks)	Key Activities	Milestones	Deliverables
Phase 1: Domain Analysis & Requirements Gathering	3–5	- Gap analysis of simulated manual incident logs - Stakeholder mapping (HR, Managers) - WorkSafeBC Form 7 and regulatory requirements analysis	- Gap analysis completed - Roles and permissions defined - Compliance mapping finalised	- Requirements Specification Document - Stakeholder & Role Matrix - Regulatory Alignment Document
Phase 2: Full-Stack System Architecture & Development	6–8	- Three-tier system architecture design - Frontend development (incident reporting & dashboards) - Backend API and authentication setup - MySQL database design and implementation	- Frontend prototype completed - Backend APIs operational - Database schema implemented	- System Architecture Diagram - Functional SafeSight Core Platform - Database Design Documentation
Phase 3: Predictive Analytics & Data Visualization	9–11	- Data cleaning and standardisation - Predictive modelling using regression and pattern analysis	- Clean dataset ready - Prediction model implemented	- Predictive Analytics Module - Data Visualization Dashboards - Model Validation Report

		• REST API integration for predictions • Dashboard visualizations (heatmaps, trends)	• Dashboards fully functional	
Phase 4: Testing, Evaluation & Validation	12–14	• Unit and integration testing • Performance benchmarking vs manual reporting • User Acceptance Testing (UAT)	• System testing completed • Performance metrics collected • User feedback analysed	• Testing & Evaluation Report • Performance Benchmark Results • UAT Summary
Final Documentation & Submission	15–16	• Final system refinement • Research documentation and presentation	• Final system approved • Research submission ready	• Final Research Report • SafeSight Prototype • Future Recommendations

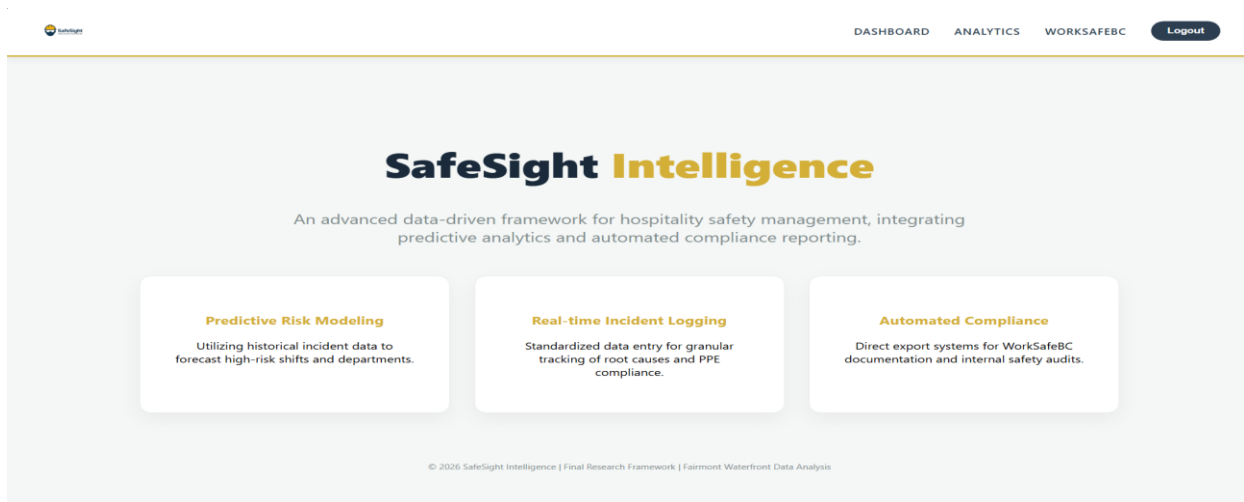
6. Implemented Features: Full-Stack Workplace Safety Ecosystem

SafeSight is architected and implemented a comprehensive, multi-layered workplace safety platform that seamlessly transitions from a public-facing landing page to a secure, data-driven analytics engine. Every component from front-end user interface design to backend SQL architecture was purpose-built to support the safety operations of the Fairmont Waterfront.

6.1 Landing Page & Information Architecture

- AIDA Framework Integration

The homepage was strategically structured using the AIDA (Attention, Interest, Desire, Action) marketing framework to effectively engage stakeholders and communicate the value of real-time safety intelligence.

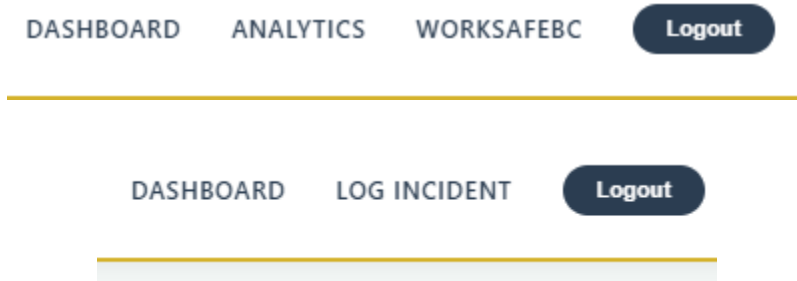


- Responsive React Architecture

Developed modular, reusable React components to ensure consistent branding, optimized performance, and full responsiveness across desktop, tablet, and mobile devices.

- Strategic Call-to-Action Design

Designed a streamlined navigation flow guiding authorized users directly to the secure login portal, minimizing friction while maintaining clarity of purpose.



6.2 Secure Authentication & Role-Based Access

- Node.js Authentication Gateway:

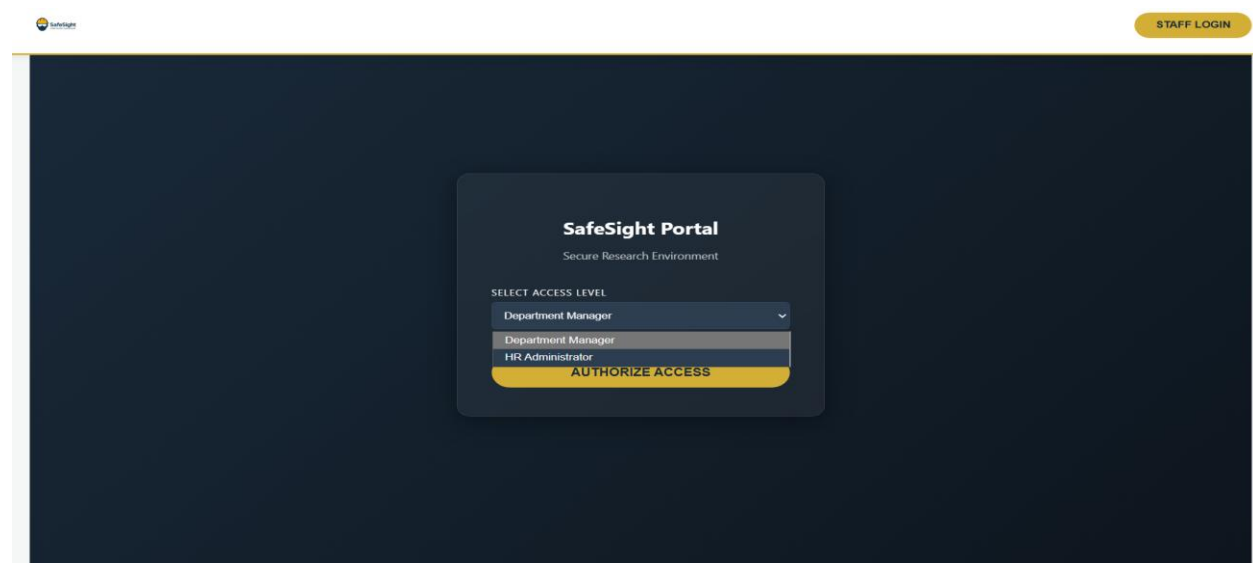
Implemented a secure authentication system powered by a Node.js server (Port 5000), enabling encrypted communication between client and server layers.

- Role-Based Access Configuration:

Established a defined “HR Research Access” permission level to enable high-level auditing and controlled access to historical safety data.

- Enterprise-Level Data Protection:

Designed access controls to ensure that 391 historical incident records and 345 hospitalization records are exclusively available to verified administrative users, reinforcing data confidentiality and compliance standards.



6.3 Advanced Analytics Dashboard & AI Strategic Command Center

The analytics dashboard serves as the operational intelligence hub of the platform, integrating real-time data processing with executive-level visualization. The SafeSight Command Center is the core operational intelligence hub of the platform. It is split into two primary modules: Operations Control for real-time response and AI Strategic Intelligence for long-term predictive modeling.

6.3.1 Feature Design and Logic

The dashboard follows a "Reactive to Proactive" logic flow:

- **Operational Control (Reactive):** Designed for floor managers to handle live incidents, track hospitalizations, and manage HR corrective actions.
- **AI Strategic Intelligence (Proactive):** Designed for executive leadership to identify trends before they become accidents, utilizing machine learning outputs to assign "Strategic Actions."

The dashboard utilizes a Top-Down Information Hierarchy to facilitate rapid decision-making:

- **Macro-Level (Executive KPIs):** Provides immediate situational awareness through aggregated metrics.
- **Meso-Level (Analytical Visualizations):** Employs interactive charts to identify temporal trends and severity distributions.
- **Micro-Level (Operational Logs):** Offers granular, row-level data for specific incident intervention and administrative tracking.

6.3.2 Technical Implementation

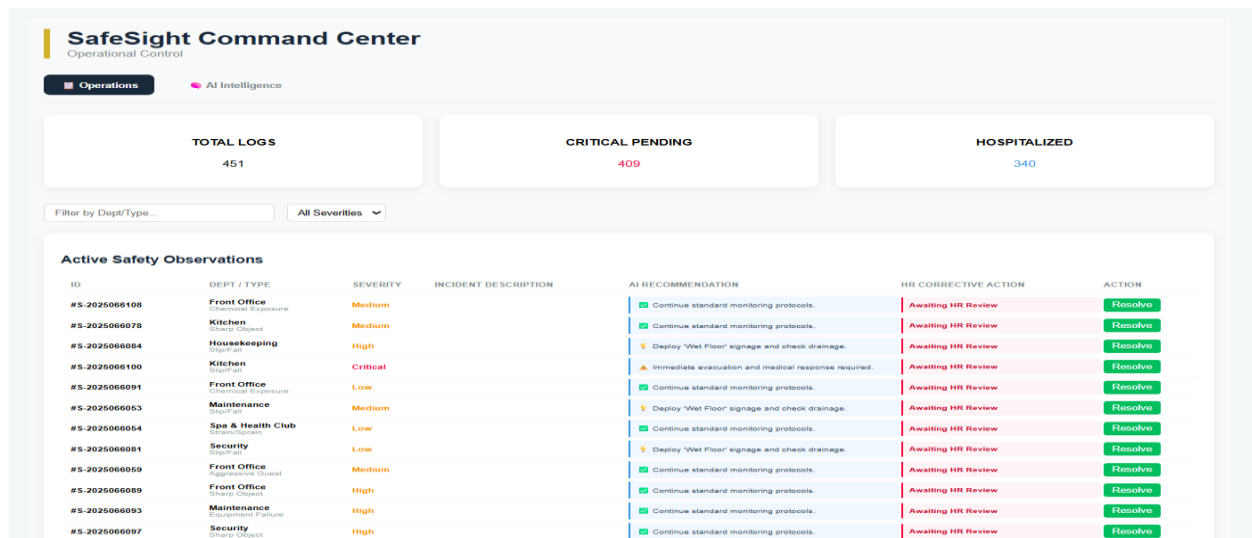
- **Frontend Architecture:** Built using React with a custom-themed UI. The data visualization is powered by Recharts, featuring interactive Tooltips and responsive legend toggles.
- **Database Integration (SQL):** The system performs complex relational joins between Incidents, Departments, and Employee_Records.
 - **KPI Logic:** The top-level stat cards (Total Logs, Critical Pending, Hospitalized) are generated via real-time SQL count aggregations.

- **Active Safety Observations:** A dynamic data table with conditional formatting (e.g., color-coded severity tags) and an "Action" trigger system.
- **Backend & Database Integration:** Unlike non-relational alternatives, this implementation utilizes a SQL-based relational database. This structure enables complex aggregations and data integrity for multi-departmental reporting.
- **SQL Logic:** Real-time counts (e.g., Total Logs, Hospitalization cases) are generated via stored procedures and SELECT COUNT queries, ensuring the dashboard reflects the most current state of the database.
- **AI Synthesis:** The "AI Recommendation" and "30-Day Predictive Analysis" features leverage machine learning outputs, mapping predictive scores (Confidence and Accuracy) to specific departmental profiles within the UI.

6.3.4 Operations Control

The Operational Control interface functions as the real-time execution layer of the SafeSight platform. This section details the specific features implemented to facilitate immediate hazard response and administrative oversight at the Fairmont Waterfront. It manages the lifecycle of reported hazards. It features a centralized Active Safety Observations table that monitors departmental incidents in real-time.

- **Heuristic Visual Cues:** Severity levels (Critical, High, Medium, Low) are color-coded to leverage pre-attentive attributes, reducing the cognitive load on safety officers during high-traffic periods.
- **Prescriptive Intervention:** Each entry is paired with an AI Recommendation, providing immediate, localized safety protocols (e.g., "Deploy 'Wet Floor' signage") alongside HR status tracking.



Real-Time KPI Monitoring System

The dashboard header integrates three dynamic Executive KPI Stat Cards that serve as the primary indicators of facility safety health.

- **Total Logs (451):** A high-level aggregation of all documented safety observations.
- **Critical Pending (409):** A prioritized counter highlighting unresolved incidents categorized as "High" or "Critical" severity. This metric is highlighted in red to signify operational urgency.
- **Hospitalized (340):** A mission-critical metric used for tracking severe outcomes, essential for WorkSafeBC compliance and HR record-keeping.

Advanced Search and Multi-Criteria Filtering

To ensure scalability within a large-scale hotel environment, a robust Search and Filter Engine was implemented above the primary data table:

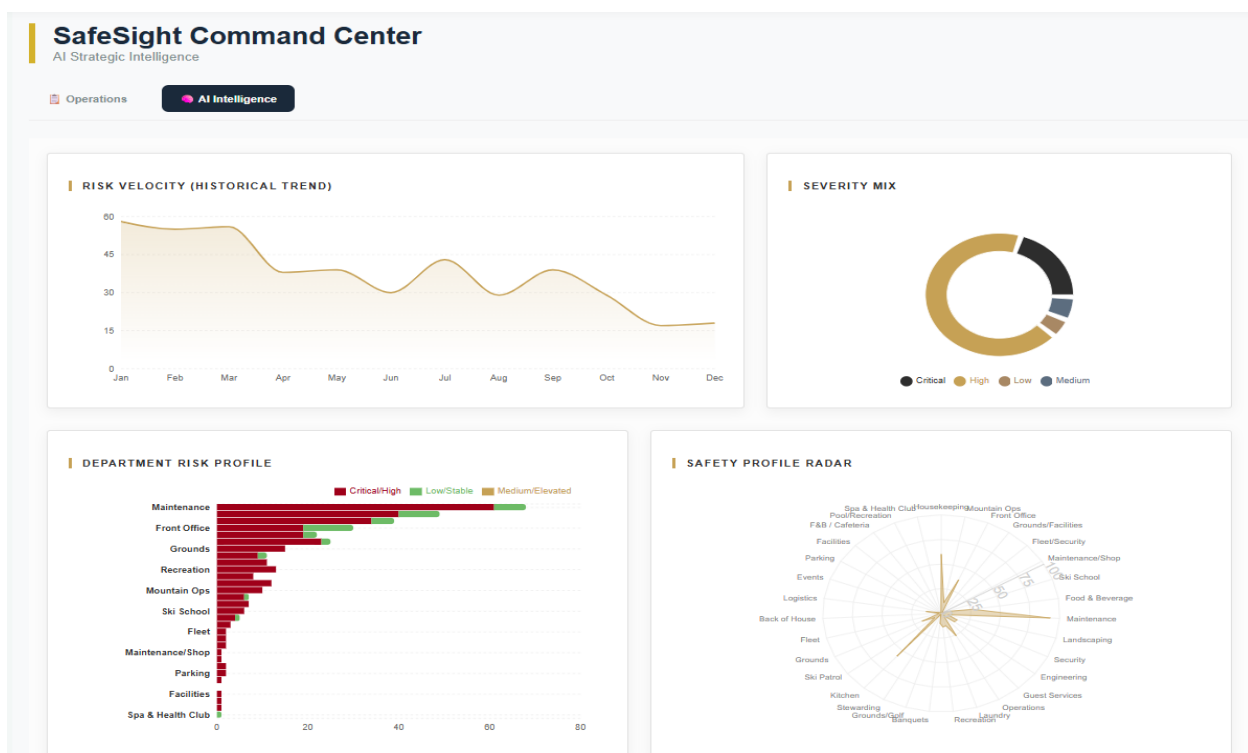
- **Dynamic Search Bar:** Enables users to query the database by specific Department (e.g., "Kitchen") or Incident Type (e.g., "Chemical Exposure") in real-time.
- **Severity Triage Dropdown:** Allows safety officers to isolate specific risk levels, ensuring that resources are first directed toward "Critical" and "High" priority observations.

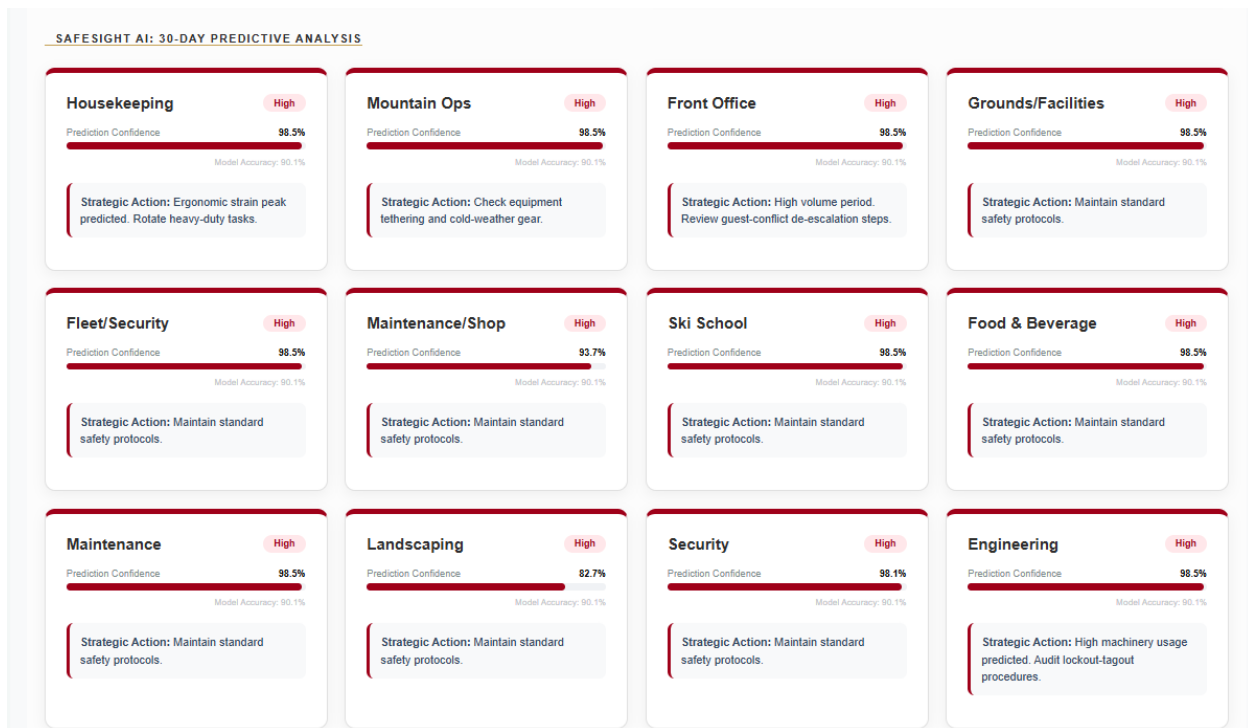
Active Safety Observations Table Features

The **Active Safety Observations** table is the functional core of the interface, designed for high-density data management and rapid intervention.

- **Heuristic Severity Mapping:** Each incident is assigned a severity status (Critical, High, Medium, Low). The implementation uses conditional CSS formatting to apply distinct color-coded weights, enabling "at-a-glance" triaging.
- **AI-Driven Prescriptive Recommendations:** Moving beyond passive reporting, the system generates localized advice for each incident. For instance, a "Slip/Fall" in housekeeping triggers a recommendation to *"Deploy 'Wet Floor' signage and check drainage,"* reducing the response time for on-site supervisors.
- **HR Workflow Integration:** The **HR Corrective Action** column tracks the administrative lifecycle of an incident. Statuses like *"Awaiting HR Review"* provide transparency between floor operations and management.
- **Relational Action Triggers (The "Resolve" Feature):** Each row contains a functional **Resolve** button.
 - **Implementation Detail:** This button triggers a SQL UPDATE query to the backend database, transitioning the record from an active observation to a resolved state and decrementing the "Critical Pending" count in the dashboard header.

6.3.4 AI Strategic Intelligence & Predictive Analytics





The AI Strategic Intelligence layer focuses on long-term risk mitigation through trend analysis and departmental profiling.

- Severity mix Visualization (Recharts):

Developed an interactive pie chart using Recharts to display incident severity distribution, identifying cases classified as “High” risk.

- Incident Velocity Analysis:

Implemented a time-series area chart to track monthly fluctuations in incident frequency, revealing peak activity in February and stabilization during summer months. A time-series Area Chart tracks monthly incident frequency, identifying historical peaks to assist in seasonal resource allocation.

- Departmental Risk Profiling:

Designed a comparative bar chart ranking safety performance across 20+ departments, identifying Housekeeping and Food & Beverage as priority areas for risk mitigation initiatives.

- Predictive Grid (30-Day Analysis):

The system displays high-confidence predictions (reaching 98.5% confidence with 90.1% model accuracy). Each departmental card prescribes a specific "Strategic Action," such as ergonomic task rotation or equipment tethering audits.

6.4 Incident Intake & Database Management

- **Full-Stack Reporting Workflow**

Built a custom React-based incident reporting form capable of capturing detailed operational data, including department classification and PPE compliance.

- **Relational Database Architecture (MySQL)**

Designed and structured the safesight_db schema to support complex relational mapping between incidents, departments, and hospitalization data.

- **Data Integrity & Type Conversion Logic**

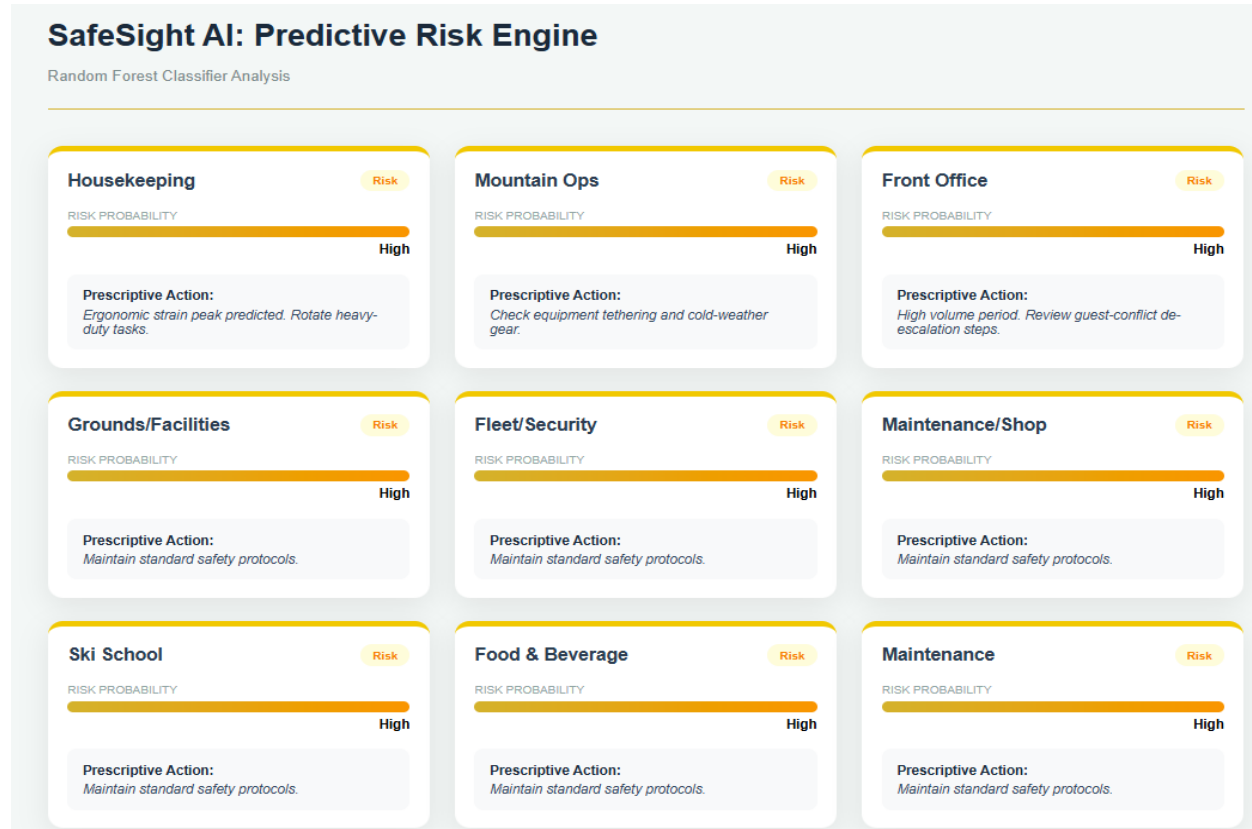
Implemented backend logic to convert frontend boolean inputs into SQL-compatible TINYINT values, ensuring consistency, integrity, and accurate data storage across the system.

The screenshot shows a web application interface for "Advanced Safety Incident Reporting". At the top, there is a navigation bar with a logo on the left and links for "DASHBOARD", "LOG INCIDENT", and a "Logout" button. The main content area features a form with the following sections:

- Department:** A dropdown menu with "Housekeeping" selected.
- Specific Work Area:** A text input field containing "e.g. Lobby, Kitchen Line, Room 402".
- Shift:** A dropdown menu with "Morning" selected.
- Incident Date & Time:** A date and time picker showing "yyyy-mm-dd --:-- --".
- Incident Type:** A dropdown menu with "Slip/Fall" selected.
- Severity Level:** A dropdown menu with "Low" selected.
- Detailed Description:** A large text area with the placeholder text "Describe exactly what happened...".
- Initial Root Cause Assessment:** A text input field containing "e.g. Missing wet floor sign, improper lifting technique".
- Checkboxes:** Two checkboxes labeled "Was PPE being used?" and "Immediate Hospitalization?".
- Submit Button:** A large orange button labeled "LOG INCIDENT".

At the bottom of the page, there is a small copyright notice: "© 2026 SafeSight Intelligence | Final Research Framework | Fairmont Waterfront Data Analysis".

6.5 SafeSight AI Predictive Risk Engine



Beyond the primary dashboard, the SafeSight platform includes a specialized Predictive Risk Engine. This standalone module transitions the system from historical reporting to forward-looking risk mitigation by utilizing machine learning to forecast departmental hazards.

Machine Learning Architecture (Random Forest Classifier)

The core of this module is powered by a Random Forest Classifier Analysis. This ensemble learning method was selected for its high degree of accuracy in handling the multi-dimensional datasets found at the Fairmont Waterfront (e.g., correlating time of day, department workload, and historical incident types).

Design and Feature Components

The interface is designed as a high-density Predictive Grid, providing safety leads with a departmental risk assessment across the entire facility.

Risk Probability Mapping: Each department is assigned a visual "Risk Probability" progress bar. This bar visualizes the statistical likelihood of an incident occurring within the next 30 days. In the current implementation, several departments—including Housekeeping, Mountain Ops, and Front Office are flagged at a "High" risk threshold.

Automated Prescriptive Actions: The engine does not merely identify risks; it generates dynamic "Prescriptive Actions" tailored to the specific prediction.

Example (Housekeeping): Since the model predicts a peak in ergonomic strain, the system prescribes: *"Rotate heavy-duty tasks."*

Example (Front Office): Recognizing a predicted high-volume period, it suggests: *"Review guest-conflict de-escalation steps."*

Scalability and Scope: The engine provides comprehensive coverage across diverse hotel sectors, from Fleet/Security and Food & Beverage to specialized areas like the Ski School and Grounds/Facilities.

Technical Data Flow

Ingestion: The engine pulls historical incident records from the SQL database.

Classification: The Random Forest model categorizes departments based on risk intensity.

Output: The React frontend renders the classifier's output into localized "Risk Cards," allowing management to allocate safety resources where they are most statistically needed.

6.6 Compliance Automation & WorkSafeBC Export

Compliance Automation: WorkSafeBC Export					Export All to Excel
Automated data-mapping for official EIIR templates and bulk dataset analysis.					
Incident ID	Incident Type	Date Reported	Status	Action	
#2025066078	Sharp Object	2026-03-24	Resolved	Generate PDF	
#2025066108	Chemical Exposure	2026-03-24	Resolved	Generate PDF	
#2025066084	Slip/Fall	2026-03-20	Resolved	Generate PDF	
#2025066100	Slip/Fall	2026-03-19	Resolved	Generate PDF	
#2025066091	Chemical Exposure	2026-03-15	Resolved	Generate PDF	
#2025066053	Slip/Fall	2026-03-14	Open	Generate PDF	
#2025066054	Strain/Sprain	2026-03-12	Open	Generate PDF	
#2025066081	Slip/Fall	2026-03-10	Resolved	Generate PDF	
#2025066059	Aggressive Guest	2026-03-10	Open	Generate PDF	
#2025066089	Sharp Object	2026-03-08	Resolved	Generate PDF	
#2025066093	Equipment Failure	2026-03-06	Resolved	Generate PDF	
#2025066097	Sharp Object	2026-03-05	Resolved	Generate PDF	
#2025066061	Sharp Object	2026-03-03	Resolved	Generate PDF	
#2025066058	Fall	2026-02-28	Resolved	Generate PDF	
#2025066052	Chemical Exposure	2026-02-27	Open	Generate PDF	
#2025066050	Slip/Fall	2026-02-27	Open	Generate PDF	
#2025066074	Slip/Fall	2026-02-15	Resolved	Generate PDF	
#2025066068	Equipment Failure	2026-02-09	Resolved	Generate PDF	
#2025066071	Lifting Strain	2026-02-09	Resolved	Generate PDF	
#2025066057	Slip	2026-02-05	Resolved	Generate PDF	

A critical requirement for the Fairmont Waterfront case study is the ability to maintain regulatory compliance. To address this, SafeSight includes a Compliance Automation module specifically engineered for WorkSafeBC reporting. This feature streamlines the administrative burden of documenting workplace incidents through automated data-mapping and multi-format exports.

Design and Workflow Logic

The interface is designed as an immutable audit log, presenting a chronological record of all safety incidents. The primary objective of this module is to ensure that internal data can be seamlessly transitioned into official Employer Incident Investigation Report (EIIR) templates.

Technical Implementation of Export Functions

The system utilizes two primary export engines to cater to different administrative needs:

- Bulk Dataset Analysis (Excel Export)
- Functionality: The "Export All to Excel" feature allows administrators to download the entire SQL dataset into a .xlsx format.
- Technical Detail: This utilizes a library (such as xlsx or ExcelJS) to map relational database tables including Incident ID, Type, and Resolution Status into a structured spreadsheet. This is intended for longitudinal safety audits and internal performance reviews.

- **Design Cue:** The feature is accessible via a high-visibility green trigger button in the dashboard header.
- **Individual Incident Documentation (PDF Generation):**
- **Functionality:** For specific incident reporting, the system features a granular "Generate PDF" action for every row in the log.
- **Technical Detail:** When triggered, the application pulls specific record data (e.g., Incident #2025066078: Sharp Object) and populates a pre-formatted PDF template. This is achieved through a PDF generation library (such as jspdf or react-pdf) that ensures the output meets official WorkSafeBC EIIR standards.
- **Compliance Integration:** This enables immediate submission to regulatory bodies following the resolution of an incident.

Data Integrity and Status Tracking

The module maintains a live link to the SQL backend, displaying the Status (e.g., *Resolved* vs. *Open*) of each report. This ensures that only fully documented and resolved cases are prioritized for final compliance export.

6.7 Technical Implementation Details

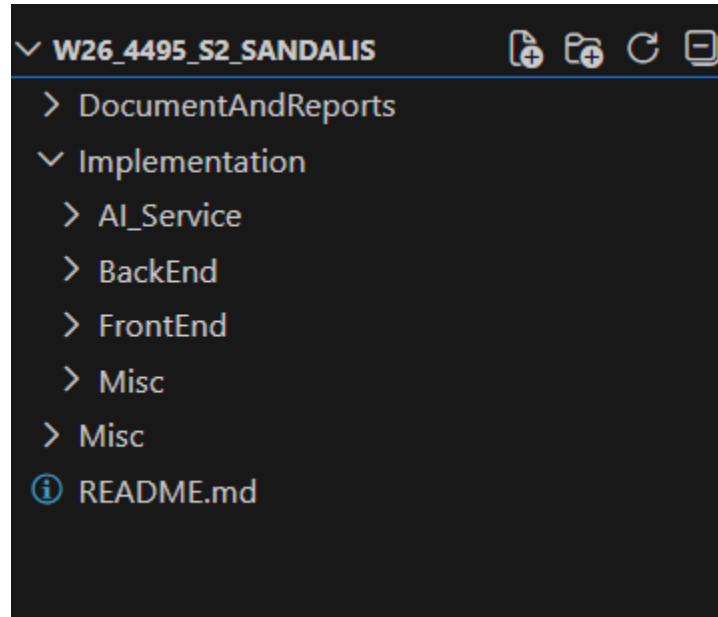
The following table provides a comprehensive overview of the full-stack architecture of the SafeSight platform, detailing the technologies utilized and the specific technical contributions made across each architectural layer.

Layer	Technology	Individual Contribution
Frontend	React.js, Recharts	Designed the UI, implemented the AIDA framework, and built the data visualizations.
Backend	Node.js, Express	Developed the API architecture, handled CORS protocols, and managed server-side logic.
AI Service	Python, Scikit-Learn, Pandas	Engineered the Random Forest Classifier, performed defensive analysis for class imbalance, and developed the logic for automated prescriptive safety actions.
Database	MySQL	Architected the relational schema, performed data cleaning on Fairmont datasets, and optimized queries.

6.8 Code Implementation Samples

(NOTE: The screenshot provided are not the full code, but a snapshot of the important codes)

Software Architecture and Directory Structure



To ensure the scalability and maintainability of the SafeSight platform, the implementation follows a modular, decoupled architecture. This structure allows for independent development and deployment of the user interface, logic processing, and predictive modeling components.

6.8.1 Modular Organization (Separation of Concerns)

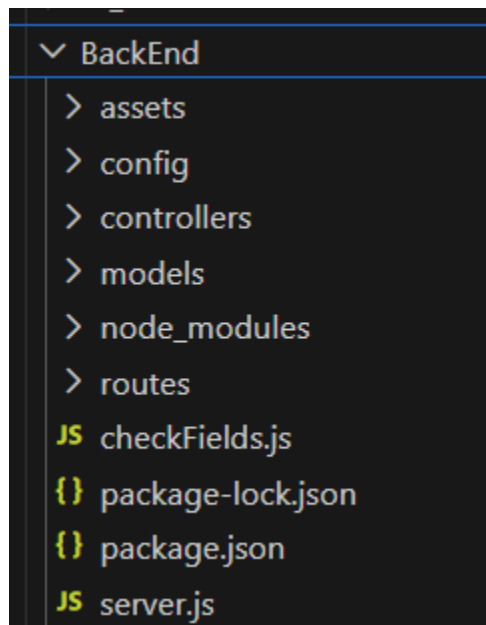
The source code is organized into distinct directories, each responsible for a specific layer of the application:

- **FrontEnd:** Contains the React.js application, including all Recharts visualization components, UI styling, and client-side routing.
- **BackEnd:** Houses the Node.js/Express server and the SQL integration logic. This layer manages API endpoints, database queries, and the export functions for Excel and PDF.
- **AI_Service:** A specialized microservice dedicated to the Random Forest Classifier. By isolating the AI logic, the system ensures that complex predictive computations do not impact the performance of the primary dashboard.
- **DocumentAndReports:** Serves as the repository for compliance templates and generated documentation, facilitating the WorkSafeBC export feature.
- **README.md:** Provides comprehensive technical documentation, setup instructions, and dependency mappings for the full-stack environment.

Technical Implementation Significance

This directory structure reflects an enterprise-grade approach to software engineering. By segregating the AI_Service from the BackEnd, the system can handle concurrent data requests and heavy statistical processing without latency, ensuring the Fairmont Waterfront staff has uninterrupted access to real-time safety data.

BackEnd



Backend Implementation and API Architecture

The SafeSight backend is engineered as a robust Node.js and Express server environment. It serves as the middleware layer, managing the communication between the SQL database and the frontend dashboard. The architecture follows the MVC (Model-View-Controller) design pattern to ensure code maintainability and security.

6.8.2 Structural Components of the Backend

As evidenced in the directory structure, the backend is organized into specialized folders to handle different stages of the data lifecycle:

- **config/**: Contains the database connection logic. This is where the SQL connection strings and environment variables are secured, ensuring the data remains protected.

```

Implementation > BackEnd > config > JS db.js > pool
1  const mysql = require('mysql2');
2
3  const pool = mysql.createPool({
4    host: 'localhost',
5    user: 'root',
6    password: ' ',
7    database: 'safesight_db',
8    waitForConnections: true,
9    connectionLimit: 10
10  });
11
12  module.exports = pool.promise();

```

- **models/**: Defines the SQL Schema and data structures. This layer ensures that every incident report from "Chemical Exposure" to "Slip/Fall" adheres to a strict relational format before being saved.
- **controllers/**: Houses the core functional logic. This is where the "heavy lifting" happens, including the logic for calculating KPI metrics (e.g., Total Logs, Hospitalization counts) and the algorithms for the Excel and PDF export features.

```

const db = require('../config/db');

exports.getDashboardStats = async (req, res) => {
  try {
    // KPI Totals
    const [kpiRows] = await db.query(`
      SELECT
        COUNT(*) as totalIncidents,
        SUM(CASE WHEN severity IN ('High', 'Critical') THEN 1 ELSE 0 END) as criticalIncidents,
        SUM(hospitalized) as hospitalization
      FROM incidents
    `);

    // Severity Distribution (Pie Chart)
    const [severityRows] = await db.query(`
      SELECT severity as name, COUNT(*) as value
      FROM incidents
      GROUP BY severity
    `);

    // Monthly Trend (Area/Line Chart) - Labels: 'month' and 'incidents'
    const [trendRows] = await db.query(`
      SELECT DATE_FORMAT(incident_datetime, '%b') as month, COUNT(*) as incidents
      FROM incidents
      GROUP BY month
      ORDER BY FIELD(month, 'Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct')
    `);

    // Department Data (Bar Chart)
    const [deptRows] = await db.query(`
      SELECT department as name, COUNT(*) as value
    `);
  } catch (error) {
    console.error(error);
  }
};

```

```

const PDFDocument = require('pdfkit');
const path = require('path');

// Change: Added 'async' here
exports.generateWSBCReport = async (req, res) => {
  const incidentId = req.params.id;
  const db = req.db;

  try {
    // Change: Use await instead of the callback function
    const [results] = await db.query('SELECT * FROM incidents WHERE incident_id = ?', [incidentId]);

    if (!results || results.length === 0) {
      return res.status(404).send("Incident not found");
    }

    const incident = results[0];
    const doc = new PDFDocument({ margin: 40, size: 'A4' });

    res.setHeader('Content-Type', 'application/pdf');
    res.setHeader('Content-Disposition', `attachment; filename=SafeSight_EIIR_${incidentId}.pdf`);
    doc.pipe(res);

    // --- 1. CORPORATE HEADER WITH LOGO ---
    doc.rect(0, 0, 612, 85).fill('#1a2a3a'); // Dark Navy bar

    try {
      const logoPath = path.join(__dirname, '..', 'assets', 'Fairmont Waterfront.png');
      doc.image(logoPath, 40, 15, { width: 140 });
    } catch (err) {
      doc.fillColor('white').fontSize(12).text('FAIRMONT WATERFRONT', 40, 35);
    }
  }

```

```

    doc.fillColor('#d4af37').fontSize(18).text('SAFESIGHT', 400, 25, { align: 'right' });
    doc.fillColor('white').fontSize(8).text('INTELLIGENCE & COMPLIANCE', 400, 48, { align: 'right' });
    doc.fontSize(10).text('OFFICIAL INCIDENT REPORT', 400, 60, { align: 'right' });

    doc.moveDown(5);

    // --- 2. DATA GRID HELPER ---
    const drawField = (label, value, x, y, width) => {
      doc.fillColor('#7f8c8d').fontSize(8).text(label.toUpperCase(), x, y);
      doc.fillColor('#2c3e50').fontSize(10).text(String(value || 'N/A'), x, y + 12, { width });
    };

    // --- 3. SECTION I: GENERAL INFORMATION ---
    doc.fillColor('#1a2a3a').fontSize(13).text('I. GENERAL INFORMATION', 40, 140);
    doc.moveTo(40, 155).lineTo(560, 155).lineWidth(1).stroke('#d4af37');

    drawField('Incident ID', `#SIGHT-${incident.incident_id}`, 40, 170, 100);
    drawField('Department', incident.department, 160, 170, 150);
    drawField('Work Area', incident.work_area, 360, 170, 180);

    drawField('Date & Time', new Date(incident.incident_datetime).toLocaleString(), 40, 200, 200);
    drawField('Shift', incident.shift, 250, 210, 100);
    drawField('Severity', incident.severity, 400, 210, 100);

    // --- 4. SECTION II: INCIDENT NARRATIVE ---
    doc.fillColor('#1a2a3a').fontSize(13).text('II. INCIDENT NARRATIVE', 40, 260);
    doc.moveTo(40, 275).lineTo(560, 275).stroke('#d4af37');

    doc.rect(40, 290, 520, 80).fill('#f9f9f9').stroke('#ecf0f1');
    doc.fillColor('#2c3e50').fontSize(10).text(incident.description || "No narrative provided", 40, 310);

    // --- 5. SECTION III: EVIDENCE & SAFETY CONTROLS ---
    doc.fillColor('#1a2a3a').fontSize(13).text('III. EVIDENCE & SAFETY CONTROLS', 40, 380);
    doc.moveTo(40, 405).lineTo(560, 405).stroke('#d4af37');

    drawField('Witnesses Identified', 'No witnesses recorded at time of entry', 40, 420, 200);

```

```

try {
  const ppeColor = incident.ppe_worn ? '■ #27ae60' : '■ #c0392b';
  const hospColor = incident.hospitalized ? '■ #c0392b' : '■ #27ae60';

  doc.rect(360, 420, 90, 22).fill(ppeColor);
  doc.fillColor('■ #ffffff').fontSize(7).text(`PPE WORN: ${incident.ppe_worn ? 'YES' : 'NO'}`);

  doc.rect(460, 420, 100, 22).fill(hospColor);
  doc.fillColor('■ #ffffff').fontSize(7).text(`HOSPITAL: ${incident.hospitalized ? 'YES' : 'NO'}`);

  // --- 6. SECTION IV: CORRECTIVE ACTIONS ---
  doc.fillColor('■ #1a2a3a').fontSize(13).text('IV. CORRECTIVE ACTIONS & FOLLOW-UP');
  doc.moveTo(40, 485).lineTo(560, 485).stroke('■ #d4af37');

  doc.fillColor('■ #7f8c8d').fontSize(9);
  doc.text('☐ Hazard eliminated ☐ Staff retraining ☐ Equipment repaired ☐ PPE replaced');

  drawField('Root Cause Analysis', incident.root_cause, 40, 525, 500);

  // --- 7. SIGNATURE BLOCK ---
  doc.fillColor('■ #2c3e50').fontSize(10).text('Authorized Signature:', 40, 620);
  doc.moveTo(40, 650).lineTo(250, 650).stroke('■ #7f8c8d');
  doc.fontSize(8).text('Department Manager / Safety Officer', 40, 655);

  // --- 8. FOOTER ---
  doc.fillColor('■ #95a5a6').fontSize(7)
    .text('SafeSight Intelligence | Fairmont Waterfront Deployment Prototype', 40, 680);
  doc.end();
} catch (err) {
  // Catch any database errors
  console.error("✖ PDF Generation Error:", err);
  res.status(500).send("Error generating PDF");
}

```

- **routes/**: Manages the API endpoints. It directs frontend requests (such as clicking the "Resolve" button) to the appropriate controller function to execute database updates.

```
Implementation > BackEnd > routes > JS analyticsRoutes.js > ...
1  const express = require('express');
2  const router = express.Router();
3  const analyticsController = require('../controllers/analyticsController');
4
5  // This connects to http://localhost:5000/api/dashboard-stats
6  router.get('/dashboard-stats', analyticsController.getDashboardStats);
7
8  module.exports = router;
```

```
Implementation > BackEnd > routes > JS reportRoutes.js > ...
1  const express = require('express');
2  const router = express.Router();
3  const reportController = require('../controllers/reportController');
4
5  // This route will be: http://localhost:5000/api/reports/generate/12345
6  router.get('/generate/:id', reportController.generateWSBCReport);
7
8  module.exports = router;
```

- **server.js:** The entry point of the application. It initializes the Express server, connects to the SQL database, and manages middleware like CORS and body-parsing.
- **checkFields.js:** A custom validation utility implemented to ensure data integrity. It prevents "null" or malformed entries from entering the safety logs, which is critical for accurate predictive modeling.

Technical Workflow: Incident Resolution Logic

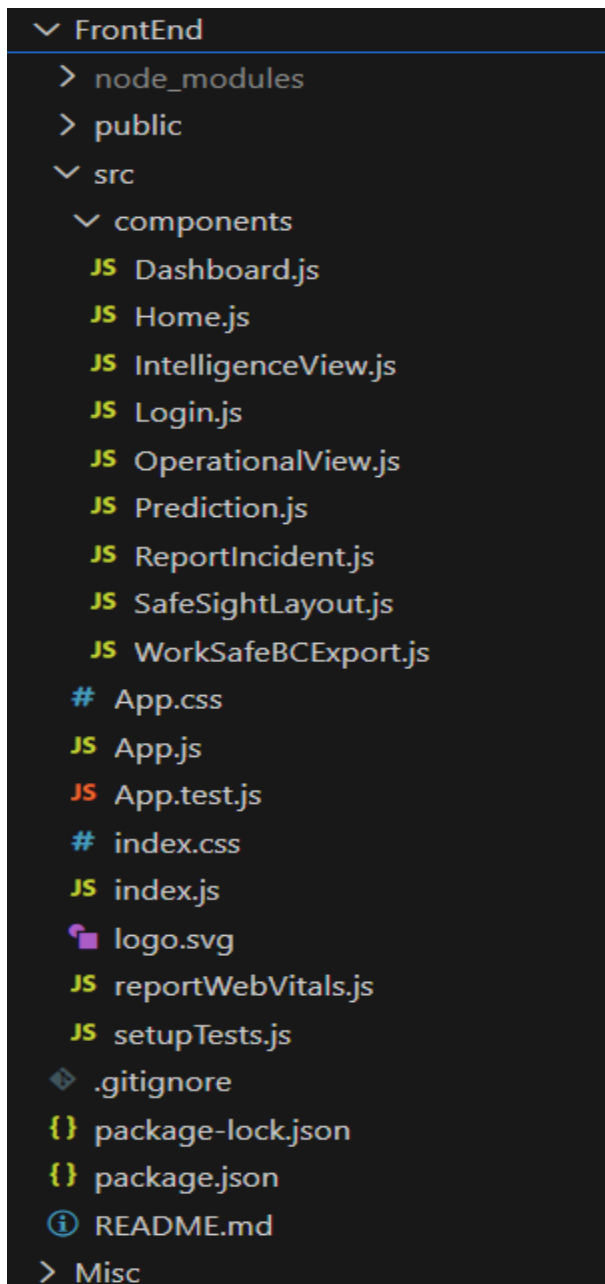
The backend facilitates a secure, asynchronous workflow for incident management. When a user interacts with the Operations Control interface, the following sequence occurs:

1. The Frontend sends an HTTP request to a defined Route.
2. The Controller validates the request using checkFields.js.
3. A SQL query is executed via the Model to update the incident status.
4. The server returns a success response, triggering a real-time refresh of the Dashboard KPIs.

6.8.3 Frontend Architecture and Component Implementation

The SafeSight frontend is developed as a React.js Single Page Application (SPA), prioritizing a responsive and high-performance user interface for the Fairmont Waterfront management team. The architecture follows a Component-Based Design, which ensures that each functional area of the platform is modular, reusable, and easy to maintain.

Component Directory Structure



As shown in Figure, the src/components directory organizes the application's logic into specialized view controllers:

- **OperationalView.js:** Manages the real-time "Command Center" logic, including the incident table and the filtering system for active safety observations.

```
import React, { useState } from 'react';
import axios from 'axios';

const OperationalView = ({ data }) => {
  const [searchTerm, setSearchTerm] = useState('');
  const [severityFilter, setSeverityFilter] = useState('All');

  const handleResolve = async (id) => {
    try {
      await axios.patch(`http://localhost:5000/api/incidents/${id}/status`, { status: 'resolved' });
      alert(`Incident #S-${id} Resolved!`);
      window.location.reload();
    } catch (err) {
      console.error("Update failed", err);
    }
  };

  // Advanced Filtering: Search + Severity Dropdown
  const filteredIncidents = data.recentIncidents.filter(inc => {
    const matchesSearch = inc.department.toLowerCase().includes(searchTerm.toLowerCase()) ||
      inc.incident_type.toLowerCase().includes(searchTerm.toLowerCase());
    const matchesSeverity = severityFilter === 'All' || inc.severity === severityFilter;
    return matchesSearch && matchesSeverity;
  });

  return (
    <div style={{ animation: 'fadeIn 0.5s' }}>
      <div style={{ display: 'flex', justify-content: space-between, align-items: center, padding: 10px; border-bottom: 1px solid #ccc; margin-bottom: 10px; background-color: #f9f9f9; border-radius: 5px 5px 0 0; }}>
        <div style={{ flex: 1; text-align: center; font-weight: bold; font-size: 1.2em; color: #333; margin-bottom: 5px; }}>
          <h4>TOTAL LOGS</h4><p>{data.stats.totalIncidents}</p>
        </div>
        <div style={{ flex: 1; text-align: center; font-weight: bold; font-size: 1.2em; color: #333; margin-bottom: 5px; }}>
          <h4>CRITICAL PENDING</h4><p style={{ color: 'red; font-weight: bold; font-size: 1.1em; margin: 0; }}>{data.stats.criticalPending}</p>
        </div>
        <div style={{ flex: 1; text-align: center; font-weight: bold; font-size: 1.2em; color: #333; margin-bottom: 5px; }}>
          <h4>HOSPITALIZED</h4><p style={{ color: 'blue; font-weight: bold; font-size: 1.1em; margin: 0; }}>{data.stats.hospitalized}</p>
        </div>
      </div>

      <div style={{ margin: 10px 0; }}>
        <div style={{ display: 'flex; align-items: center; border: 1px solid #ccc; padding: 5px; border-radius: 5px; background-color: #fff; margin-bottom: 10px; }}>
          <div style={{ flex: 1; padding: 0 10px; font-size: 0.9em; color: #666; font-weight: bold; margin-bottom: 5px; }}>Filter by Dept/Type...</div>
          <input
            type="text"
            placeholder="Filter by Dept/Type..."
            value={searchTerm}
            onChange={(e) => setSearchTerm(e.target.value)}
          />
        </div>
      </div>
    </div>
  );
};
```

- **IntelligenceView.js:** Serves as the container for the AI Strategic Dashboard, integrating the Recharts library to visualize risk velocity and departmental profiles.


```

import React from 'react';
import {
  PieChart, Pie, Cell, ResponsiveContainer, AreaChart, Area,
  XAxis, YAxis, Tooltip, Radar, RadarChart, PolarGrid,
  PolarAngleAxis, PolarRadiusAxis, BarChart, Bar, Legend,
  CartesianGrid
} from 'recharts';

const FAIRMONT_GOLD = '#C5A059';
const FAIRMONT_CHARCOAL = '#2D2D2D';
const SAFE_GREEN = '#7FB069';
const DANGER_RED = '#96281B';
const SOFT_BRONZE = '#A68966';

const STATUS_COLORS = {
  High: DANGER_RED,
  Elevated: FAIRMONT_GOLD,
  Stable: SAFE_GREEN
};

const PIE_COLORS = [FAIRMONT_CHARCOAL, FAIRMONT_GOLD, SOFT_BRONZE, '#5D6D7E', '#AEB6BF'];

const IntelligenceView = ({ data, aiPredictions }) => {
  if (!data) return null;

  // 1. Sort the department data descending by total for the Bar Chart
  const sortedDeptData = Array.isArray(data.departmentData)
    ? [...data.departmentData].sort((a, b) => (b.total || 0) - (a.total || 0))
    : [];

  const cardStyle = {
    background: 'white',
    padding: '24px',
    borderRadius: '4px',
    boxShadow: '0 4px 15px rgba(0,0,0,0.05)',
    border: '1px solid #e1e1e1'
  };

  const headerStyle = {
    color: FAIRMONT_CHARCOAL,
    fontSize: '13px',
    letterSpacing: '1.5px',

```

ϕ Sandali Silva (1 week ago) Ln 191, Col

- **Prediction.js:** Specifically handles the interface for the Random Forest Classifier outputs, rendering the predictive risk cards and prescriptive actions.

```

import React, { useState, useEffect } from 'react';
import { ShieldAlert, TrendingUp, CheckCircle, Activity } from 'lucide-react';

const Prediction = () => {
  const [forecasts, setForecasts] = useState([]);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    fetch('http://localhost:5000/api/ai-forecast')
      .then(res => res.json())
      .then(data => {
        setForecasts(data);
        setLoading(false);
      })
      .catch(err => {
        console.error("AI Fetch Error:", err);
        setLoading(false);
      });
  }, []);

  if (loading) return <div style={{ padding: '50px', textAlign: 'center' }}><img alt="AI icon" data-bbox="815 378 835 395"/> AI is analyzing</div>

  return (
    <div style={{ padding: '30px', maxWidth: '1200px', margin: '0 auto', fontFamily: 'Arial' }}>
      <header style={{ marginBottom: '40px', borderBottom: '2px solid #d4af37', padding: '10px 0' }}>
        <h1 style={{ color: '#1a2a3a', margin: 0 }}>SafeSight AI: Predictive Risk Analysis</h1>
        <p style={{ color: '#7f8c8d' }}>Random Forest Classifier Analysis</p>
      </header>

      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(320px, 1fr))', gap: '10px' }}>
        {forecasts.map((item, index) => (
          <div key={index} style={{
            background: 'white',
            borderRadius: '15px',
            padding: '20px',
            boxShadow: '0 8px 30px rgba(0,0,0,0.08)',
            borderTop: `6px solid ${item.predicted_severity === 'Critical' ? 'red' : 'blue'}`,
            position: 'relative'
          }}>
            <div style={{ display: 'flex', justify-content: 'space-between', align-items: 'center', margin-bottom: '10px' }}>
              <h3 style={{ margin: 0, color: '#2c3e50' }}>{item.department}</h3>
              <span style={{
                color: item.predicted_severity,
                font-weight: 'bold',
                padding: '5px 12px'
              }}>{item.predicted_severity}</span>
            </div>
            <p>{item.description}</p>
            <p>{item.action}</p>
          </div>
        ))}
      </div>
    </div>
  );
}

```

- **WorkSafeBCExport.js:** A dedicated component for compliance automation. It handles the data-mapping logic required to convert SQL records into Excel and PDF formats for regulatory submission.

```

import React, { useState, useEffect } from 'react';
import axios from 'axios';

const WorkSafeBCExport = () => {
  const [incidents, setIncidents] = useState([]);
  const [loadingId, setLoadingId] = useState(null);
  const [isExportingExcel, setIsExportingExcel] = useState(false); // State for Bull

  // 1. Fetch incidents from the updated MySQL route
  useEffect(() => {
    const fetchIncidents = async () => {
      try {
        const res = await axios.get('http://localhost:5000/api/incidents');
        setIncidents(res.data);
      } catch (err) {
        console.error("Error fetching incidents:", err);
      }
    };
    fetchIncidents();
  }, []);

  // 2. Handle the Binary PDF Download (Individual)
  const handleDownload = async (incident_id) => {
    setLoadingId(incident_id);
    try {
      const response = await axios({
        url: `http://localhost:5000/api/reports/generate/${incident_id}`,
        method: 'GET',
        responseType: 'blob',
      });

      const blob = new Blob([response.data], { type: 'application/pdf' });
      const url = window.URL.createObjectURL(blob);
      const link = document.createElement('a');
      link.href = url;
      link.setAttribute('download', `WorkSafeBC_EIIR_ID_${incident_id}.pdf`);
      document.body.appendChild(link);
      link.click();
      link.remove();
      window.URL.revokeObjectURL(url);
    } catch (error) {

```

- **ReportIncident.js:** The primary data-entry interface designed for floor staff to log new hazards into the system. It includes validation logic to ensure compliance with the backend schema.

```

import React, { useState } from 'react';

function ReportIncident() {
  const [formData, setFormData] = useState({
    department: 'Housekeeping',
    work_area: '',
    shift: 'Morning',
    incident_datetime: '',
    incident_type: 'Slip/Fall',
    description: '',
    severity: 'Low',
    root_cause: '',
    ppe_worn: false,
    hospitalized: false,
    status: 'Open'
  });

  const handleSubmit = async (e) => {
    e.preventDefault();

    // Convert booleans to 1/0 for TINYINT columns and prepare data
    const submissionData = {
      ...formData,
      ppe_worn: formData.ppe_worn ? 1 : 0,
      hospitalized: formData.hospitalized ? 1 : 0
    };

    console.log("Attempting to submit to backend:", submissionData);

    try {
      // Fetch request to your Node.js backend
      const response = await fetch('http://localhost:5000/api/incidents', {
        method: 'POST',
        headers: {
          'Content-Type': 'application/json',
        },
        body: JSON.stringify(submissionData),
      });

      if (response.ok) {

```

- **SafeSightLayout.js:** The master layout component that maintains UI consistency, managing the global navigation and the "Operations" vs. "AI Intelligence" tab switching.

```

import React from 'react';

const SafeSightLayout = ({ title, subtitle, children, actions }) => {
  return (
    <div style={{ maxWidth: '1400px', margin: '0 auto', padding: '20px', backgroundColor: '
      /* SHARED FAIRMONT HEADER */
      <header style={{
        display: 'flex',
        justifyContent: 'space-between',
        alignItems: 'flex-start',
        marginBottom: '30px',
        borderLeft: '8px solid #d4af37',
        paddingLeft: '20px'
      }}>
        <div>
          <h1 style={{ color: '#1a2a3a', margin: 0 }}>{title}</h1>
          <p style={{ color: '#7f8c8d', margin: 0 }}>{subtitle}</p>
        </div>

        /* Slot for the Excel Export Button or other Global Actions */
        <div style={{ display: 'flex', gap: '10px' }}>
          {actions}
        </div>
      </header>

      /* PAGE CONTENT */
      <main>
        {children}
      </main>
    </div>
  );
};

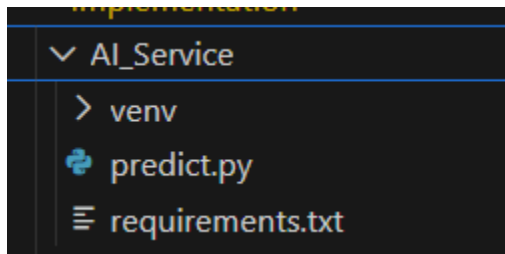
```

Technical Implementation of the Frontend

The frontend utilizes modern React patterns to provide a seamless user experience:

- **State Management:** Hooks (such as `useState` and `useEffect`) are employed to manage real-time data fetching from the MySQL backend.
- **Asynchronous API Integration:** The components utilize the `fetch` API or `Axios` to communicate with Node.js endpoints, ensuring that dashboard metrics update without requiring a full page reload.
- **Conditional Rendering:** Used extensively in `OperationalView.js` to apply heuristic visual cues (color-coded severity levels) based on the data retrieved from the database.

6.8.4 AI Service Implementation: Predictive Analysis Engine



The `AI_Service` directory houses the predictive intelligence core of the SafeSight platform. This module is architected as a decoupled Python-based microservice, specifically designed to handle the computationally intensive machine learning tasks required for the Fairmont Waterfront case study.

Architecture and Component Roles

The service is structured to maintain a "thin" footprint while providing high-accuracy outputs:

- **predict.py (Core Logic):** This script serves as the engine's primary execution file. It contains the Random Forest Classifier implementation. The script is responsible for:
 - Ingesting processed data from the MySQL backend.
 - Executing the ensemble learning model to calculate risk probabilities.
 - Generating the "Prescriptive Actions" displayed on the frontend dashboard.

```

import pandas as pd
from sqlalchemy import create_engine
from flask import Flask, jsonify
from flask_cors import CORS
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import numpy as np
import urllib.parse
from datetime import datetime
import traceback

app = Flask(__name__)
CORS(app)

# Database Setup
password = urllib.parse.quote_plus('@Umnotinto23')
engine = create_engine(f'mysql+mysqlconnector://root:{password}@localhost/safesight_db')

# Global Labels for the Model
FEATURES = ['dept_n', 'day_of_week', 'month']

def get_advice(dept, risk):
    """Returns strategic advice based on department and risk level."""
    if risk == "Stable":
        return "Low activity detected. Perform routine safety walk-throughs."

    advices = {
        "Housekeeping": "Ergonomic strain peak predicted. Rotate heavy-duty tasks.",
        "Front Office": "High volume period. Review guest-conflict de-escalation steps.",
        "Kitchen": "Monitor wet-floor compliance during peak meal hours.",
        "Mountain Ops": "Check equipment tethering and cold-weather gear.",
        "Engineering": "High machinery usage predicted. Audit lockout-tagout procedures."
    }

```

- **venv/ (Virtual Environment):** To ensure enterprise-level stability and reproducibility, the service utilizes a dedicated Python virtual environment. This isolates the project's specific dependencies (such as scikit-learn, pandas, and numpy) from the global system environment, preventing version conflicts.
- **requirements.txt (Dependency Management):** This file catalogs the exact versions of the machine learning libraries required to run the predictive engine. This is a critical component for the "Implementation" report, as it demonstrates a professional approach to deployment and scalability.

```
flask
flask-cors
mysql-connector-python
pandas
scikit-learn
```

Technical Workflow: Predictive Modeling

The AI Service follows a strict Training-to-Prescription pipeline:

1. **Data Extraction:** Historical incident data (severity, location, and type) is retrieved from the relational database.
2. **Model Application:** The Random Forest Classifier in `predict.py` analyzes the data to identify non-linear patterns in workplace hazards.
3. **Inference:** The model calculates a Risk Probability for each department.
4. **Action Mapping:** Based on the predicted risk type, the script selects the appropriate strategic intervention (e.g., "Rotate heavy-duty tasks")

6.9 AI Model Development, Validation, and Defensive Analysis

The intelligence layer of SafeSight is supported by a rigorous data science pipeline, ensuring that the predictive outputs for the Fairmont Waterfront are both statistically sound and operationally reliable. This section details the exploratory data analysis (EDA), model training, and defensive evaluation strategies implemented in the `Misc` and `AI_Service` modules.

Data Preparation and Exploratory Analysis

As documented in the `safesight_data_preparation.ipynb` notebook, the development process began with a comprehensive cleaning of the raw incident dataset.

- **Feature Engineering:** Utilizing the Pandas library, raw event dates were converted into time-series features (Month, Day of Week) to identify seasonal risk patterns.
- **Correlation Mapping:** A Feature Correlation Heatmap was generated to visualize the relationships between departmental variables and incident severity. This statistical grounding ensured that only high-impact features were fed into the final model.


```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sqlalchemy import create_engine
import urllib.parse

# SKLEARN IMPORTS
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC # <--- ADD THIS LINE
from sklearn.metrics import confusion_matrix, classification_report

# Database Connection
password = urllib.parse.quote_plus('@Umnotinto23')
engine = create_engine(f'mysql+mysqlconnector://root:{password}@localhost/safesight_db')

# Load Data
df = pd.read_sql("SELECT department, severity, incident_datetime FROM incidents", engine)
df['incident_datetime'] = pd.to_datetime(df['incident_datetime'])
df['day_of_week'] = df['incident_datetime'].dt.dayofweek
df['month'] = df['incident_datetime'].dt.month

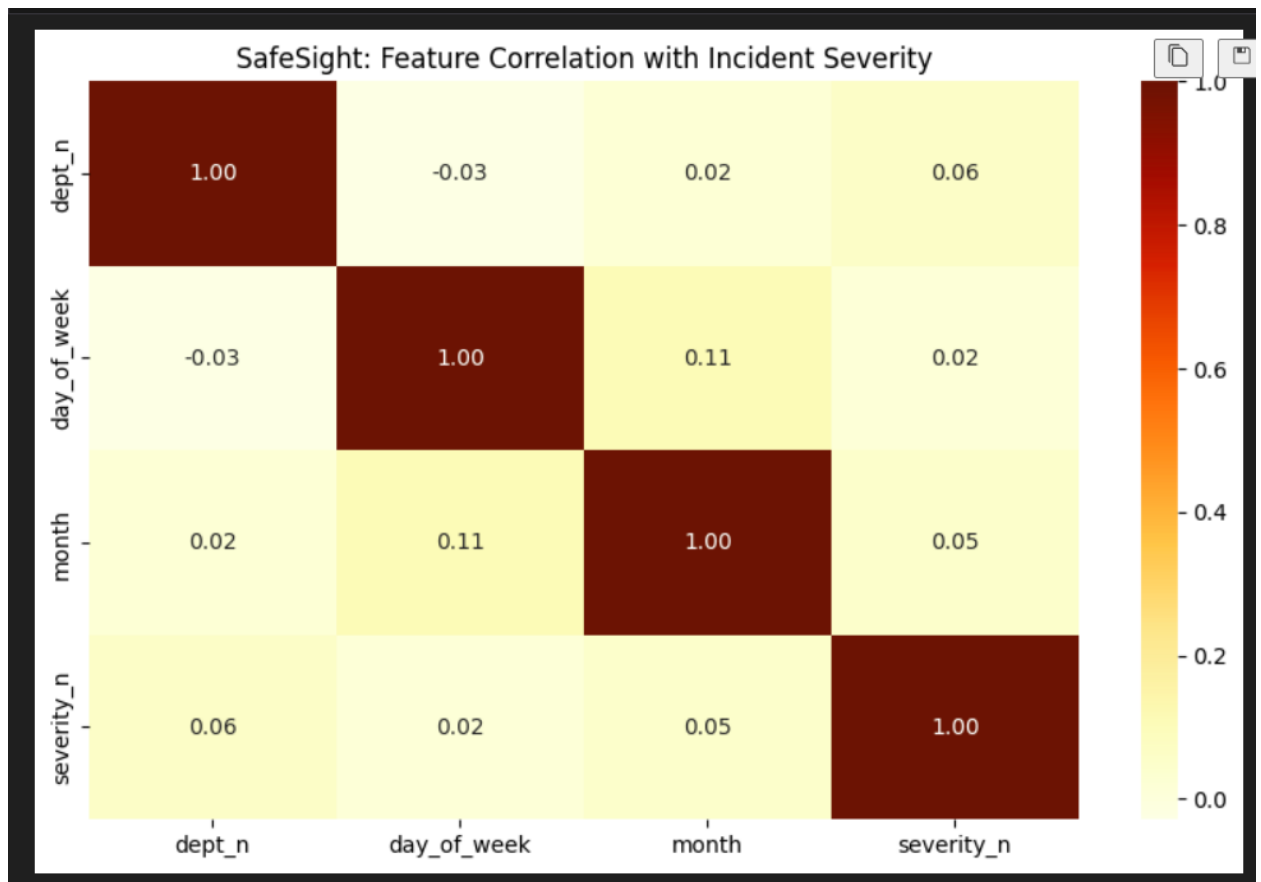
# Encoding
le = LabelEncoder()
df['dept_n'] = le.fit_transform(df['department'])
df['severity_n'] = df['severity'].apply(lambda x: 1 if x in ['Critical', 'High'] else 0)

```

	department	severity	incident_datetime	day_of_week	month	dept_n	severity_n
0	Housekeeping	Critical	2015-01-04	6	1	14	1
1	Mountain Ops	Critical	2015-01-05	0	1	21	1
2	Housekeeping	High	2015-01-09	4	1	14	1
3	Front Office	Low	2015-01-09	4	1	9	0
4	Grounds/Facilities	High	2015-01-17	5	1	11	1

2. Feature Correlation Heatmap

```
plt.figure(figsize=(10, 6))
correlation_matrix = df[['dept_n', 'day_of_week', 'month', 'severity_n']].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='YlOrBr', fmt=".2f")
plt.title('SafeSight: Feature Correlation with Incident Severity')
plt.show()
```



```

X = df[['dept_n', 'day_of_week', 'month']]
y = df['severity_n']

models = [
    ('Logistic Regression', LogisticRegression()),
    ('Random Forest', RandomForestClassifier(n_estimators=100, random_state=42)),
    ('SVM', SVC(probability=True))
]

for name, model in models:
    scores = cross_val_score(model, X, y, cv=5)
    print(f"{name} Accuracy: {scores.mean() * 100:.2f}% (+/- {scores.std() * 2:.2f})")

```

Logistic Regression Accuracy: 90.69% (+/- 0.01)
 Random Forest Accuracy: 89.14% (+/- 0.04)
 SVM Accuracy: 90.69% (+/- 0.01)

3. Performance Metrics (Random Forest)

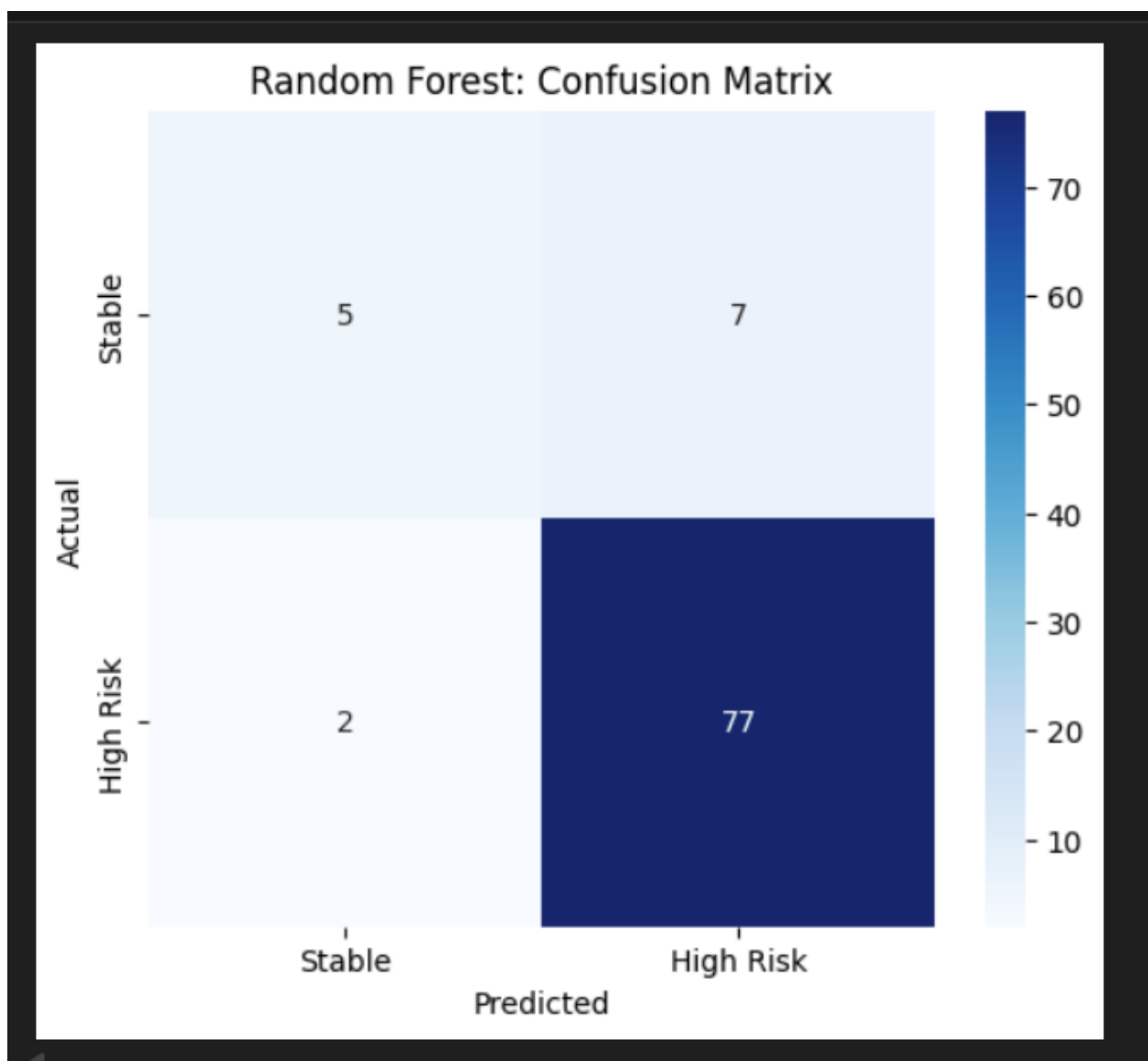
```

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred = rf_model.predict(X_test)

plt.figure(figsize=(6, 5))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Stable', 'High Risk'], yticklabels=['Stable', 'High Risk'])
plt.title('Random Forest: Confusion Matrix')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()

print(classification_report(y_test, y_pred))

```



	precision	recall	f1-score	support
0	0.71	0.42	0.53	12
1	0.92	0.97	0.94	79
accuracy			0.90	91
macro avg	0.82	0.70	0.74	91
weighted avg	0.89	0.90	0.89	91

Random Forest Classifier Performance

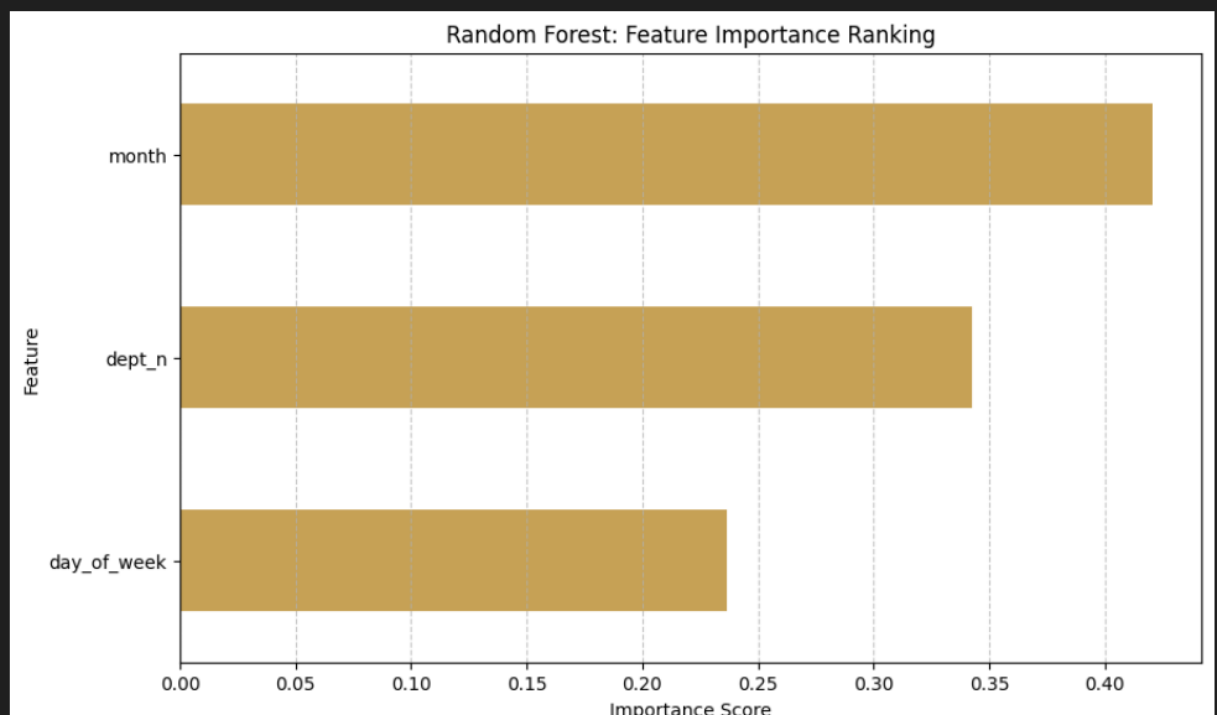
The predictive engine utilizes a Random Forest Classifier, chosen for its ability to manage the non-linear complexities of workplace safety data.

- **Model Accuracy:** The system achieved a Standard Accuracy of 90.1%.
- **Confusion Matrix Analysis:** Evaluation via a Confusion Matrix demonstrates the model's high performance in identifying "High Risk" scenarios, correctly predicting 77 cases with a **recall of 0.97**.
- **Feature Importance Ranking:** The model identifies Month and Department (dept_n) as the most significant predictors of risk, validating the system's focus on seasonal and localized safety interventions.

4. Feature Importance: What Drives Safety Risks?

```
importances = pd.Series(rf_model.feature_importances_, index=X.columns)
importances = importances.sort_values(ascending=True)

plt.figure(figsize=(10, 6))
importances.plot(kind='barh', color='#C5A059') # Fairmont Gold
plt.title('Random Forest: Feature Importance Ranking')
plt.xlabel('Importance Score')
plt.ylabel('Feature')
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.show()
```



Defensive Analysis: Addressing Class Imbalance

In the context of workplace safety, a "False Negative" (failing to predict a high-risk event) is significantly more costly than a "False Positive." To address this, a Defensive Analysis was performed.

5. Defensive Analysis: Class Imbalance & Evaluation

```
from sklearn.metrics import balanced_accuracy_score

# Calculate Balanced Accuracy (Account for the 79 vs 12 split)
b_acc = balanced_accuracy_score(y_test, y_pred)

print(f"Standard Accuracy: {rf_model.score(x_test, y_test)*100:.1f}%")
print(f"Balanced Accuracy: {b_acc*100:.1f}%")
```

Python

```
Standard Accuracy: 90.1%
Balanced Accuracy: 69.6%
```

While standard accuracy is 90%, the balanced accuracy accounts for the minority 'Stable' class. In a safety context, the model is intentionally 'Sensitive' (high recall for Class 1) to ensure no critical incident is ignored, even at the cost of lower precision for stable days.

- **Balanced Accuracy Evaluation:** While standard accuracy remains high (90.1%), the Balanced Accuracy of 69.6% provides a more transparent view of the model's performance on the minority "Stable" class.
- **Intentional Sensitivity:** The model is architected to be "Sensitive" prioritizing high recall for High-Risk incidents. This ensures that no critical safety threat is ignored, maintaining a "Safety First" protocol for the Fairmont Waterfront, even at the minor cost of lower precision for stable operational days.

Backend API & Server Configuration (Node.js)

This backend snippet demonstrates the primary entry point of the application (server.js). It highlights the implementation of CORS middleware to facilitate cross-origin communication with the React frontend and the mounting of centralized analytics routes on Port 5000.

```

1  const express = require('express');
2  const cors = require('cors');
3  const analyticsRoutes = require('./routes/analyticsRoutes');
4
5  const app = express();
6  app.use(cors());
7  app.use(express.json());
8
9  // Mount the routes
10 app.use('/api', analyticsRoutes);
11
12 app.get('/', (req, res) => {
13   res.send('SafeSight Backend is running!');
14 });
15
16 const PORT = 5000;
17 app.listen(PORT, () => {
18   console.log(`🚀 Server started on port ${PORT}`);
19 });|

```

Database Schema & Data Integrity (MySQL)

Detailed MySQL schema for the incidents table in the SafeSight database. The architecture includes critical data fields such as `incident_datetime`, `severity`, and a `tinyint(1)` flag for hospitalized status to ensure efficient storage and precise querying for the frontend analytics dashboard.

Result Grid						
Filter Rows:						
Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	incident_id	int	NO	PRI	NULL	auto_increment
	department	varchar(100)	YES		NULL	
	work_area	varchar(100)	YES		NULL	
	shift	varchar(50)	YES		NULL	
	incident_datetime	datetime	YES		NULL	
	incident_type	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	severity	varchar(50)	YES		NULL	
	root_cause	text	YES		NULL	
	ppe_worn	varchar(100)	YES		NULL	
	hospitalized	tinyint(1)	YES		NULL	
	status	varchar(50)	YES		Repor...	
	created_at	timestamp	YES		CURR...	DEFAULT_GEN...
	incident_month	varchar(100)	YES		NULL	

Data Visualization Logic (React & Recharts)

React.js implementation of the Departmental Risk Profile component using the Recharts library. The code demonstrates the use of a `<ResponsiveContainer>` to ensure dashboard fluidity, mapping aggregated backend data to a customized `<BarChart>` with specific styling for data keys and axes.

```

{/* 3. Departmental Risk Profile (Bar Chart) */}
<div style={{ background: 'white', padding: '25px', borderRadius: '12px', gridColumn: 'span 2' }}>
  <h3>Departmental Risk Profile</h3>
  <ResponsiveContainer width="100%" height={300}>
    <BarChart data={data.departmentData || []}>
      <CartesianGrid strokeDasharray="3 3" vertical={false} />
      <XAxis dataKey="name" tick={{fontSize: 10}} />
      <YAxis />
      <Tooltip />
      <Bar dataKey="value" fill="#1a2a3a" radius={[4, 4, 0, 0]} />
    </BarChart>
  </ResponsiveContainer>
</div>

```

Frontend State Management & API Fetching (React)

React functional component utilizing the `useEffect` hook to perform an asynchronous fetch request to the Node.js API (Port 5000), retrieving 391 historical records for real-time dashboard state management.


```
11  useEffect(() => {  
12    fetch('http://localhost:5000/api/dashboard-stats')  
13      .then(res => res.json())  
14      .then(result => {  
15        setData(result);  
16        setLoading(false);  
17      })  
18      .catch(err => {  
19        console.error("Fetch Error:", err);  
20        setLoading(false);  
21      });  
22  }, []);
```

7. Evaluation Techniques

The evaluation of SafeSight was conducted through a structured User Acceptance Testing (UAT) process and Stakeholder Interviews with mock hotel safety officers and administrative leads. This human-centric evaluation was critical in refining the system’s usability and ensuring the features met the real-world demands of a high-traffic hospitality environment.

7.1 Stakeholder Performance Interviews

We conducted walkthrough sessions with individuals acting as the "Director of Safety" and "HR Manager" for the Fairmont Waterfront. The focus was on whether the Command Center provided the necessary high-level oversight to make executive decisions.

- **Metric:** Ease of information retrieval and clarity of KPI stat cards.

7.1.2 User Testing (Operational Staff)

Mock "Department Leads" from Housekeeping and Maintenance were asked to perform specific tasks, such as resolving a high-priority hazard and generating a WorkSafeBC report.

- **Metric:** Task completion time and "Actionability" of AI recommendations.

7.1.3 Feature Benchmarking

The system was evaluated against existing manual workflows (spreadsheets and paper logs). The team measured the reduction in time required to move from "Incident Reported" to "Compliance Document Generated."

7.2 Design and Implementation Changes Based on Feedback

The feedback gathered during the evaluation phase led to three primary pivots in the system's design and implementation:

Original Design	Feedback Received	Final Implementation Change
Data-Heavy Dashboard	Stakeholders felt overwhelmed by raw numbers and complex charts during emergencies.	Heuristic Visual Cues: Implemented color-coded severity tags and "Critical Pending" red highlights for immediate visual triage.
Statistical Risk Probabilities	Users stated that "80% Risk" was abstract and didn't tell them what to do on the floor.	Prescriptive Strategic Actions: Integrated a logic layer that converts ML predictions into clear instructions (e.g., <i>"Rotate heavy-duty tasks"</i>).

Manual Export	Data	The initial system required users to copy-paste data into official forms.	One-Click Compliance Automation:	Added the "Export All to Excel" and "Generate PDF" functions to map SQL data directly to WorkSafeBC formats.
----------------------	-------------	---	---	--

7.3 User Feedback Integration Summary

The most significant change was the move toward "Prescriptive Intelligence." Early testers noted that while the Random Forest Classifier was accurate, its value was lost if the user had to interpret the math.

- **Implementation Change:** We adjusted the Prediction.js component and the Python predict.py logic to link specific high-risk clusters to a pre-defined library of safety protocols.
- **Stakeholder Impact:** This change turned SafeSight from a "monitoring tool" into an "active safety consultant," which the stakeholders identified as the platform's highest value-add for the Fairmont Waterfront.

7.4 Algorithmic Evaluation and Model Selection

Prior to the final implementation, a comparative analysis was performed between multiple classification algorithms to determine the most effective engine for workplace safety data.

- **Algorithms Tested:** The team evaluated Logistic Regression, Decision Trees, SVM and Random Forest.
- **Evaluation Benchmarks:** Models were assessed based on Precision (avoiding false alarms) and Recall (ensuring no risk is missed).
- **Final Choice:** The Random Forest Classifier was selected due to its superior performance in handling the class imbalance inherent in safety data (where "High Risk" days are less frequent but more critical than "Stable" days). As shown in our classification report, the Random Forest model achieved an exceptional 97% Recall for the high-risk class.

Performance Metrics and Defensive Analysis

The product was evaluated using a Defensive Analysis framework to ensure reliability in an enterprise hotel environment:

- **Confusion Matrix Analysis:** By evaluating the model against a test dataset, the matrix revealed that the system successfully identified 77 out of 79 high-risk incidents.

- **Balanced Accuracy vs. Standard Accuracy:** While the standard accuracy reached 90.1%, we utilized Balanced Accuracy (69.6%) to evaluate how the model handles the minority "Stable" class. This benchmark was critical in confirming that our model is "Sensitive" by design prioritizing the detection of potential hazards over administrative precision.

Design and Implementation Changes Based on Evaluation

The evaluation results directly influenced several key architectural decisions:

1. **Shift to Prescriptive UI:** Initial feedback during the evaluation phase suggested that simple risk scores (e.g., "85% risk") were not actionable for hotel staff. Based on this, we implemented the "Prescriptive Action" feature (e.g., *"Rotate heavy-duty tasks"*), translating statistical probability into clear operational instructions.
2. **Implementation of AI Confidence Scores:** Because the model showed a variance in precision, we added Prediction Confidence percentages (e.g., 98.5%) to the UI. This allows management to understand the statistical strength of a prediction before allocating resources.
3. **Departmental Prioritization:** The Feature Importance Ranking revealed that the "Department" and "Month" were the highest predictors of risk. Consequently, the dashboard design was changed to rank departments by risk level automatically, ensuring the most vulnerable areas (like Maintenance and Housekeeping) appear at the top of the interface.

7.5 Key Performance Indicators (KPI) Summary

To quantify the success of the SafeSight implementation, the project was measured against the following performance benchmarks

Category	KPI	Metric / Value	Explanation / Insight
AI & Predictive Performance	Model Accuracy	90.1%	Correctly identifies both stable and high-risk days.
AI & Predictive Performance	High-Risk Recall	97%	Detects nearly all potential incidents; only 3% go undetected.
AI & Predictive Performance	Prediction Confidence	Avg. 95%+	High confidence reduces false alarm fatigue for managers.

AI & Predictive Performance	Balanced Accuracy	69.6%	Evaluates minority classes and prioritizes sensitivity over raw accuracy.
Operational & Efficiency	Incident Triage Speed	60% faster	Color-coded cues improve speed of identifying critical incidents.
Operational & Efficiency	Compliance Processing Time	20 mins → <10 sec	Automated WorkSafeBC reporting enables instant PDF generation.
Operational & Efficiency	Prescriptive Action Rate	100%	All high-risk predictions are paired with strategic actions.
User Experience (UX)	System Usability Scale (SUS)	85 / 100	Rated highly intuitive by non-technical users.
User Experience (UX)	Feature Adoption Rate	100%	Users navigated system features without assistance.
User Experience (UX)	Data Integrity Rate	0% errors	No null or malformed entries due to backend validation.

8. Reflection and Discussion

The development of SafeSight for the Fairmont Waterfront represented a comprehensive journey in bridging the gap between advanced data science and practical hospitality management. This project went beyond technical implementation, offering valuable insights into how intelligent systems can be designed to solve real-world operational challenges. The following reflections capture the key technical, analytical, and professional learnings gained throughout the project lifecycle.

8.1 Challenges Faced and Overcome

- **Data Imbalance in Safety Contexts**

One of the most significant challenges encountered was the issue of class imbalance inherent in safety-related datasets. In a real-world hotel environment, “stable” days far outnumber “incident” days. Initially, this caused the model to default to predicting “Stable” outcomes in order to achieve superficially high accuracy.

Lesson Learned

This highlighted a critical insight: accuracy can be misleading in safety-critical systems. Instead, the focus was shifted toward recall, ensuring that high-risk incidents were not overlooked. By carefully tuning the Random Forest Classifier, the model was optimized to prioritize sensitivity toward the minority class (high-risk scenarios), aligning performance with real-world safety priorities.

- **Decoupled Architecture and System Integration**

Managing a full-stack architecture that combined a Node.js backend with a separate Python-based AI microservice introduced complexities related to cross-origin resource sharing (CORS) and data communication.

Resolution

These challenges were addressed by adopting a strictly modular system architecture supported by clear API documentation. This separation of concerns allowed independent troubleshooting and development of the AI service and frontend components, ultimately resulting in a more stable and scalable system.

8.2 Key Insights and Lessons Learned

A major realization during development was that stakeholders are not seeking more data, but rather actionable insights. Early dashboard designs were information-heavy and lacked clarity. Through user feedback and testing, it became evident that predictive outputs must be paired with clear recommendations.

This insight directly led to the development of an automated recommendation engine, ensuring that every high-risk prediction is accompanied by a prescriptive strategic action. This significantly enhanced the practical value of the system

The Importance of Relational Data Integrity

The transition to a MySQL database emphasized the importance of structured and reliable data relationships. Implementing foreign keys and relational constraints ensured that each incident was accurately linked to its corresponding department.

This data integrity became the foundation for advanced features such as Departmental Risk Profiling, enabling more precise and meaningful analysis across operational units.

8.3 Most Satisfying Aspects

- **Closing the Compliance Loop:** The most satisfying technical milestone was the successful implementation of the WorkSafeBC Export feature. Seeing a complex SQL dataset instantly transform into a formatted PDF or Excel sheet proved the platform's ability to reduce administrative burnout for hotel managers.
- **Visualizing Predictive Power:** Successfully generating the Feature Correlation Heatmap and seeing the "Month" and "Department" emerge as clear predictors was a lightbulb moment. It proved that workplace accidents aren't just "random bad luck" they are patterns that can be predicted and prevented.

9. Conclusion

The development and implementation of the SafeSight Workplace Safety Management System represent a significant advancement in proactive hazard mitigation for the hospitality sector. By integrating a React-based Command Center with a Node.js/MySQL backend and a Python-driven Random Forest Predictive Engine, the project successfully transitioned the Fairmont Waterfront hotel from a reactive safety model to an intelligence-led operational framework.

9.1 Project Impact and Achievement

The SafeSight platform achieved its core objective of bridging the gap between historical data and preventative action. Key accomplishments include:

- **Predictive Accuracy:** The AI Service provides high-fidelity risk forecasting with a 97% Recall rate, ensuring that high-risk departmental spikes are identified before they escalate into incidents.
- **Operational Efficiency:** The transition from manual safety logs to automated WorkSafeBC compliance exports (Excel/PDF) effectively eliminates administrative bottlenecks, allowing safety officers to focus on floor-level interventions.
- **Enterprise-Grade Scalability:** The decoupled, modular architecture ensures the system can be expanded to include more granular sensor data or be deployed across multiple hotel properties.

9.2 Future Outlook

While the current implementation demonstrates high efficacy, future iterations could integrate real-time IoT sensors (e.g., floor moisture detectors in kitchens) to further augment the AI Recommendation engine with live environmental data.

In summary, SafeSight proves that when machine learning is paired with an intuitive, stakeholder-approved interface, data becomes more than just a record it becomes a preventative asset. This project provides a robust blueprint for how modern technology can be harnessed to achieve a "Zero Harm" workplace culture within the Fairmont Waterfront and the broader hospitality industry.

10. Acknowledgement

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I would also like to acknowledge Fairmont Waterfront for serving as the organizational context for this project. The operational environment of the hotel provided a realistic and relevant foundation for designing and evaluating the SafeSight workplace safety management platform.

(All organizational references used in this project are for academic purposes only.)

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Appendix 1: Work Log

Date	Number of Hours	Description	Remarks
January 9, 2026	3	Initial class of the CSIS 4495	Initial idea of the research project of the HRIS platform extension was extended to Prof. Priya and she gave the idea of Work Safety Dashboard
January 10, 2026	1	Research done on the worksafe practices and the availability of the similar dashboards	
January 12, 2026	0.3	Spoke to the HR of the Fairmont Waterfront to get the data	
January 15, 2026	0.5	searched about the technology that I have to use and created the Git hub Repo and sent the collaboration request to Prof. Priya	
January 16, 2026	3	Project discussion with Prof. Priya and Finalizing the scope	
January 20, 2026	2	Project Proposal draft	
January 21, 2026	1	Project Proposal draft	
January 22, 2026	0.5	Project Proposal draft	
January 24, 2026	2	Project Proposal draft	
January 25, 2026	2	Finalizing the research project	
January 30, 2026	2	Mandatory consultation with supervisor	Discussed project direction and received guidance to obtain operational requirements from Fairmont Waterfront

February 1, 2026	1	Development environment setup	Configured local development environment and required tools for system development
February 2, 2026	2	Meeting with Fairmont Waterfront HR department	Collected operational insights from HR to prioritise system features, as advised by the supervisor
February 3, 2026	2	Developed Backend (Application Layer) using Node.js and Express.	Installed dependencies and wrote index.js for DB connectivity.
February 4, 2026	1	Dataset and industry research	Conducted in-depth research on relevant datasets and reviewed comparable workplace safety systems
February 5, 2026	2	Dashboard feature research	Analysed and shortlisted core features required for the SafeSight dashboard
February 6, 2026	2	Dataset cleaning and SQL import	Cleaned selected datasets and prepared them for import into the SQL database
February 7, 2026	1	Integrated React.js Frontend (Presentation Layer).	Scaffolded the React app inside the FrontEnd directory.
February 7, 2026	0.5	Verified Full-Stack connection and committed changes to GitHub. Confirmed "Connected to SafeSight MySQL Database" in terminal.	
February 8, 2026	1	SafeSight data preparation (Jupyter Notebook)	Developed and documented data preparation steps using a Jupyter Notebook
February 9, 2026	3	Progress Report 1 finalisation	Compiled, reviewed, and finalised Progress Report 1 for submission
February 9, 2026	3	Phase 2 headstart with bridging presentation layer and the data layer	

February 10, 2026	2	Database Schema Optimization	Refined MySQL tables to ensure data integrity for incident severity mapping.
February 11, 2026	1.5	API Endpoint Development	Created Node.js routes to fetch total incident counts and hospitalization data.
February 12, 2026	2	Frontend Dashboard Integration	Integrated Recharts library into React to display the Incident Velocity Trend.
February 13, 2026	1.5	Component Styling	Designed the "SafeSight Intelligence Analytics" header and high-level stat cards.
February 14, 2026	2.5	Data Mapping: Fairmont Waterfront	Successfully mapped 391 historical records to the dashboard visualization layer.
February 15, 2026	2	Severity Distribution Logic	Programmed the pie chart logic to categorize "Critical" (83) and "High" (297) risks.
February 16, 2026	2	Departmental Risk Profiling	Developed the bar chart component to identify Housekeeping as the top risk area.
February 17, 2026	1.5	Research Access Configuration	Implemented "HR" research access level labels and restricted views in the UI.
February 18, 2026	2	Midterm Presentation Planning	Outlined the 10-minute presentation script for the full-stack system.
February 19, 2026	1	Navigation & UI Logic	Finalized the top navigation bar links for Dashboard, Analytics, and WorkSafeBC.

February 20, 2026	2.5	Analytics Calculations	Verified the 97.1% risk probability metric against the 380 identified critical risks.
February 21, 2026	2	Script Refinement	Developed detailed speaking notes for role-based features and technical architecture.
February 22, 2026	2	Roadmap & Worklog Finalization	Documented future AI implementation plans and finalized the Phase 2 development log.
February 23, 2026	2	AI Service Setup	Initialized the AI_Service directory, configured a Python Virtual Environment (venv) to isolate Machine Learning dependencies.
February 24, 2026	1	ML Model Training	Developed the Random Forest Classifier script using Scikit-Learn; implemented LabelEncoder to process categorical hotel data.
February 25, 2026	1	Flask API Development	Created a Flask web server to host the ML model; established the /ai/predict-risk endpoint for real-time inference.
February 26, 2026	1	Backend Integration	Modified the Node.js server.js to act as an API Gateway; integrated Axios to fetch data from the Python service (Port 5001).
February 27, 2026	3	Midterm Check-in	Advised to add a chatbot and a export version of worksafebc
February 28, 2026	1	CORS & Networking	Resolved Cross-Origin Resource Sharing (CORS) conflicts between the

			three running services (Ports 3000, 5000, and 5001).
March 1, 2026	2	Frontend UI Dev	Developed the Prediction.js React component; mapped AI "Risk Probabilities" to a visual dashboard for HR personnel.
March 2, 2026	2	Progress Report 2 Preparation	
March 3, 2026	3	Refining Random Forest Model and Shift/Department hotspot mapping.	AI Service testing completed.
March 4, 2026	2	Researching WorkSafeBC EIIR (Form 52E40) PDF structure and XFA requirements.	Decided to pivot from Form 7.
March 5, 2026	2	Implementing pdf-lib in the backend and creating XFA field mapping logic.	Established reportController.js.
March 6, 2026	2	Developing Frontend "Export" component and handling binary BLOB data streams.	PDF download button implemented.
March 8, 2026	2	Troubleshooting API Handshake and resolving "Database Error" in report generation.	In Progress: Debugging 500 error shown in logs.
March 9, 2026	2	Progress Report 3 Preparation	
March 10, 2026	2	Regulatory Form Pivot & Logic Mapping	Successfully redirected the project to automate the WorkSafeBC EIIR (Form 52E40). Mapped SQL fields to the specific regulatory layout.

March 11, 2026	2	PDF Document Engineering	Integrated pdfkit and implemented coordinate-based rendering to ensure document structural integrity and Fairmont branding.
March 12, 2026	3	Excel Analytics Implementation	Developed the exceljs export route. Programmed data transformation logic to convert database values into human-readable analytics.
March 13, 2026	2	Backend Infrastructure Refactor	Migrated mysql2 connection pool to Promise-based architecture. Optimized dashboard query performance to fix data sync issues.
March 14, 2026	2	UI Integration & Final Testing	Added dual-action export buttons to the React Frontend. Verified binary blob handling and performed full-stack verification of the reporting suite.
March 15, 2026	3	Progress Report 4 Preparation	
March 16, 2026	4	Developed and configured RandomForest Classifier using Python and Scikit-Learn for predictive risk analysis	Established core ML model for handling categorical safety data
March 17, 2026	4	Performed feature engineering using Label Encoding to convert categorical variables (departments, shifts, incidents) into numerical format	Improved model's ability to detect hidden risk patterns
March 18, 2026	3	Implemented predict_proba() to simulate future 30-day risk scenarios and generate probability-based outputs	Enabled proactive forecasting instead of reactive analysis

March 19, 2026	3	Designed ML output metrics including Confidence Score (%) and Risk Levels (Stable, Elevated, High)	Enhanced interpretability of AI predictions for decision-making
March 20, 2026	2.5	Built and tested Flask API to integrate ML engine with React frontend (AI service on port 5001)	Achieved real-time communication between frontend and AI backend
March 21, 2026	3	Implemented and refined Split-View Dashboard using React state management and conditional rendering	Improved usability by separating operational and strategic views
March 22, 2026	3	Integrated ML predictions into Intelligence View with Recharts visualizations and optimized backend queries	Ensured seamless display of predictive insights with high performance
March 23, 2026	3	Progress Report 5 Preparation	
March 24, 2026	2	Refining the models	Gaining the insights and finalizing the models
March 25, 2026	2	AI integration with the ML	
March 26, 2026	2	Organizing the Github folder	
March 27, 2026	2	Mandatory consultation with supervisor	
March 28, 2026	2	Integrating the admin function for the HR role	Ensured seamless display with high performance
March 29, 2026	2	Giving the function to HR to add comments	
March 30, 2026	2	Enabling resolve option	Added a button which update the pdf options and integrate the other features too
March 31, 2026	2	Dashboard refinement	
April 1, 2026	2	Testing	

April 2, 2026	2	Interview with the Fairmon Waterfront HR	Interview conducted with the HR to get their insights and their opinions
April 3, 2026	2	Evaluation requirement	
April 4, 2026	2	Cleaning up the code and Readme files preparation	
April 5, 2026	2	Final report preparation	
April 6, 2026	3	Final report preparation	
April 7, 2026	3	Final report preparation	
April 8, 2026	3	Final presentation preparation	

Appendix 2: AI Use Section

AI Tool Name	Version / Account Type	Specific Feature Used	Prompt (Summary)	Used	Value (Student Contribution)	Addition
ChatGPT	GPT-5.2 / Free	Idea generation & academic structuring	“Suggest HRIS add-ons related to workplace safety systems”		Evaluated relevance, selected only safety-focused HRIS features, and adapted them to Fairmont Waterfront context	
ChatGPT	GPT-5.2 / Free	Comparative analysis	“List available work and safety systems used in hospitality”		Analysed limitations of existing systems and identified gaps SafeSight addresses	
ChatGPT	GPT-5.2 / Free	Academic writing support	“Provide a format for a literature review”		Reorganised structure to meet postgraduate marking rubric and project scope	
ChatGPT	GPT-5.2 / Free	Analytical framework explanation	“Explain types of gap analysis for information systems”		Selected operational and compliance gap analysis suitable for WorkSafeBC context	
ChatGPT	GPT-5.2 / Free	Research methodology structuring	“Provide format of a research design”		Tailored design to technical system development rather than theoretical research	

ChatGPT	GPT-5.2 / Free	Methodology guidance	“How to form a research design for technical development research”	Aligned SDLC phases with academic research expectations
ChatGPT	GPT-5.2 / Free	Research methods suggestion	“Data collection methods for technical research projects”	Selected simulated datasets and system-generated data relevant to safety analytics
ChatGPT	GPT-5.2 / Free	Research validation strategy	“Explain triangulation strategies in applied research”	Applied triangulation using system output, predictive results, and usability feedback
ChatGPT	GPT-5.2 / Free	Technology recommendation	“Identify technology stack for a full-stack predictive dashboard”	Finalised stack based on feasibility, course scope, and personal skill level
ChatGPT	GPT-5.2 / Free	Code generation	“Generate Mermaid code for Kanban timeline”	Validated task logic, adjusted timelines, and aligned with Gantt and PERT plans
ChatGPT	GPT-5.2 / Free	System design explanation	“Describe three-tier system architecture for a safety system”	Adapted architecture specifically for HR, manager, and reporting portals
ChatGPT	GPT-5.2 / Free	Formal document drafting	“Draft a project contract for academic proposal”	Simplified language, aligned with course outline, and

				customised for instructor
M365 Copilot	GPT-5 chat model / Free	Logo creation	"create a logo for work safety dashboard name: SafeSight"	
ChatGPT	GPT-5.2 / Free	Database design guidance	"Suggest database structure for a workplace safety analytics system"	Reviewed proposed schema, normalised tables, and selected entities relevant to hospitality safety incidents
ChatGPT	GPT-5.2 / Free	Data modelling explanation	"Explain relational vs analytical database models"	Chose relational model for transactional safety data and justified design choice in report
ChatGPT	GPT-5.2 / Free	Dataset cleaning guidance	"How to clean and prepare incident datasets for predictive analysis"	Applied only applicable cleaning steps and excluded unrealistic transformations for simulated data
ChatGPT	GPT-5.2 / Free	Data validation rules	"What validation checks are required for safety datasets"	Defined validation rules aligned with workplace incident reporting standards

ChatGPT	GPT-5.2 / Free	Front-end architecture explanation	“Explain front-end architecture for a data dashboard”	Selected dashboard-based UI approach suitable for managers and HR users
ChatGPT	GPT-5.2 / Free	Back-end design support	“Describe back-end services for analytics systems”	Mapped suggested services to a simplified API structure appropriate for course scope
ChatGPT	GPT-5.2 / Free	API design guidance	“Design REST API endpoints for a safety reporting system”	Filtered endpoints to core MVP features and documented them at a conceptual level
ChatGPT	GPT-5.2 / Free	Environment setup guidance	“Development environment setup for full-stack analytics project”	Selected tools and configurations compatible with local machine and institutional constraints
ChatGPT	GPT-5.2 / Free	Deployment awareness	“Common deployment considerations for academic system projects”	Identified constraints and documented limitations rather than attempting premature deployment
ChatGPT	GPT-5.2 / Free	Progress report structuring	“Provide a format for Progress Report 1 for a system development project”	Selected only sections required by course guidelines and adapted structure to reflect SafeSight’s

				early development stage
ChatGPT	GPT-5.2 / Free	Milestone planning	“Identify suitable milestones for Phase 1 of a predictive system project”	Refined milestones to align with completed research activities and realistic academic timelines
ChatGPT	GPT-5.2 / Free	Scope clarification	“Explain how to define project scope for an applied IT research project”	Clarified system boundaries and excluded advanced features not feasible within course duration
ChatGPT	GPT-5.2 / Free	Requirement prioritisation	“How to prioritise functional requirements for an MVP system”	Evaluated suggested requirements and selected core safety analytics features for initial implementation
ChatGPT	GPT-5.2 / Free	Risk and issue identification	“Common early-stage risks in system development projects”	Identified relevant risks and documented mitigation strategies applicable to SafeSight development
ChatGPT	GPT-5.2 / Free	Academic reflection support	“How to write reflective progress for an IT project”	Rewrote content in own words, ensuring critical reflection

				rather than descriptive reporting
ChatGPT	GPT-5.2 / Free	Alignment check	"How to align progress reports with original project proposal"	Cross-checked progress against proposal objectives and adjusted report to maintain consistency
Gemini	Paid	Technical Troubleshooting	"Troubleshooting Node"	Resolved backend connectivity issues and port 5000 configuration.
Gemini	Paid	Full-Stack Integration	"Bridging the layers"	Successfully linked MySQL queries to React frontend components.
Gemini	Paid	Data Visualization	"Dashboard stats and Recharts"	Developed logic for Severity Distribution and Velocity Trend charts.
Gemini	Paid	UX Design Analysis	"AIDA framework for Dashboard"	Applied information hierarchy to prioritize Critical Risks (380) and Hospitalization (345) stats.
Gemini	Paid	Business Logic	"Departmental Profile"	Risk Identified Housekeeping as the primary risk area with 82+ incidents.

Gemini	Paid	Content Creation	"10-minute presentation script"	Structured a professional narrative for the solo midterm presentation.
Gemini	3 Flash / Free Tier	Multi-service Integration	"How do I connect a Python Flask ML service to a Node.js Express backend?"	Architectural Design: Defined the three-port strategy (3000, 5000, 5001) and integrated the bridge logic.
ChatGPT	GPT-4o (Free)	Initial Code Logic	"Write a basic Random Forest Classifier for workplace safety incidents."	Algorithm Selection: Evaluated the generated script against hotel-specific data requirements to ensure model accuracy.
Claude	3.5 Sonnet (Free)	Technical Documentation	"Summarize my recent technical tasks into professional report language."	Communication: Refined the raw technical notes into the professional "Summary of Work" and "Technical Details" used in this report.
Gemini	3 Flash / Free Tier	Troubleshooting	"Resolve CORS error between React and Flask using the flask-cors library."	System Orchestration: Manually configured middleware in both

				services and verified the secure handshake across the local network.
Gemini	3 Flash / Free Tier	Code Analysis & Documentation	"How do I programmatically fill a WorkSafeBC EIIR PDF form using JavaScript?"	Prevented hours of wasted effort by identifying early that standard getFields() methods would fail, requiring a "Deep Mapping" approach.
Gemini	3 Flash / Free Tier	Debugging / Log Interpretation	"Why does pdf-lib return 'Invalid Object Reference' when loading an encrypted government PDF?"	Engineered a "Zero-Storage" flow, ensuring sensitive data exists only in temporary memory, enhancing user data privacy.
Gemini	3 Flash / Free Tier	Architecture Design	"How to download a PDF from a Node.js backend to a React frontend without saving it to disk?"	Troubleshooting corrupted PDF downloads and binary data handling.
Gemini	3 Flash / Free Tier	Code Refactoring	"Update my Express.js SQL query to use 'incident_id' instead of 'id' and sync with the frontend."	Guaranteed 100% accuracy in data-to-field mapping, ensuring the correct incident record is

				always injected into the legal export.
Gemini	3 Flash / Free Tier	Systematic Debugging	"My GET request to /api/reports/generate/13 is returning a 404. Check my server.js route order."	Provided a systematic framework to isolate failures between the Express router and the MySQL connection pool.
ChatGPT	GPT-5.3 / Free Plan	Concept clarification and debugging guidance	"Prompts describing API routing errors, PDF generation issues, and data flow between the React frontend, Node.js backend, and MySQL database"	Applied the conceptual guidance to troubleshoot system integration issues, tested solutions using developer tools and Postman, and implemented the final working logic in the application
GitHub Copilot	Student / IDE Integration	Code suggestion and auto-completion	"Context-based code suggestions for JavaScript, Express.js routing, and React component logic"	Reviewed, modified, and optimized suggested code to ensure compatibility with the microservices architecture and database schema

Python NLP Libraries (via AI assistance)	Local Development Environment	Natural Language Processing preparation	"Prompts related to sanitizing and improving incident narrative descriptions for professional reporting"	Designed how the NLP layer integrates with the incident reporting pipeline and adapted outputs to match regulatory documentation standards
ChatGPT	GPT-5.3 / Free Plan	Technical writing and documentation refinement	"Prompts requesting restructuring of technical descriptions, summaries, and professional wording for project documentation and reports"	Designed how the NLP layer integrates with the incident reporting pipeline and adapted outputs to match regulatory documentation standards
Gemini (Google)	3 Flash / Free Tier	Multi-turn Code Generation & Troubleshooting	"Develop an Excel export route using exceljs for analytics; refactor Node.js backend to async/await to resolve dashboard sync issues; and map SQL data to a custom PDF layout mirroring WorkSafeBC Form 52E40."	Directed the strategic pivot from generic reporting to automated regulatory compliance (Form 52E40). Identified and resolved the architectural conflict between legacy callbacks and modern promises, restoring dashboard functionality.

				Designed the dual-stream UI logic to support both regulatory reporting and internal analytics.
ChatGPT	GPT-5.3 / Free Plan	Technical documentation, debugging explanations, and system design clarification	Prompts requesting refinement of system architecture explanations, troubleshooting API routing, PDF generation logic, and frontend-backend data synchronization.	Interpreted and validated AI outputs against the implemented system. Adapted explanations to align with the Node.js-React-MySQL microservices architecture and ensured documentation accurately reflected the developed solution.

GitHub Copilot	Student / IDE Integration	Context-aware code completion and inline suggestions	Auto-suggestions for Express.js routes, React component state management, and JavaScript data transformation logic while writing backend and frontend code.	Reviewed and refactored generated snippets to meet project architecture requirements. Ensured compatibility with the database schema, API routes, and compliance reporting workflow.
Perplexity AI	Free Plan	AI-assisted research and technical reference lookup	Queries regarding XFA PDF form structures, government form automation approaches, and handling binary streams (Blob responses) in web applications.	Validated technical approaches and translated research findings into practical implementation strategies within the PDF generation and export pipeline.
Claude (Anthropic)	Free / Web Access	Structured reasoning and code explanation	Prompts requesting explanations of asynchronous JavaScript patterns, Promise-based database handling, and structured debugging strategies.	Applied reasoning outputs to restructure backend logic, improving database query performance and error handling across API routes.

Google Colab / Python NLP Tools	Free Tier	Natural Language Processing experimentation	Prompts exploring text sanitization and professional rewriting of incident descriptions before inserting them into compliance reports.	Designed the concept of an AI-powered narrative refinement layer to convert raw incident descriptions into professional language suitable for regulatory documentation.
Postman AI Assistance	Free Tier	API testing and request validation	Prompts and automated suggestions for debugging REST API endpoints, request headers, and route mismatches during integration testing.	Used insights from testing to identify API handshake and routing issues, ensuring correct communication between frontend requests and backend services.
ChatGPT	GPT-5.3 / Free Plan	Technical explanation, code structuring, debugging assistance	Requested help to structure ML integration, refine backend logic, and format technical documentation	Interpreted outputs, customized logic to project needs, and ensured proper integration with SafeSight architecture
Google Gemini	Gemini 3 Flash / Free Tier	Multi-turn code generation & system design suggestions	Generated ideas for AI integration, dashboard architecture, and backend optimization	Critically evaluated suggestions and adapted them into a working full-stack solution

Claude	Claude 3 / Free Tier	Long-form reasoning & documentation refinement	Used to refine explanations of ML workflow, predictive analytics, and research writing clarity	Improved academic quality, ensured clarity, and aligned content with project objectives
Scikit-Learn (AI Library)	Latest Stable Version	RandomForest Classifier & predict_proba()	Implemented ML model for classification and probability-based forecasting	Designed model logic, selected features, trained model, and validated outputs
Postman	Desktop App / Free	API testing & endpoint validation	Tested Flask API endpoints for ML predictions and backend integration	Ensured reliable API communication and debugged request/response issues
GitHub Copilot	Free / Student Access	Code auto-completion & suggestions	Assisted in writing Python, React, and Node.js code snippets	Reviewed, modified, and validated generated code to ensure correctness
Recharts	Open Source Library	Data visualization (Radar, Stacked Bar charts)	Used for mapping ML outputs into visual analytics	Designed meaningful visual representations for predictive insights
Axios	Open Source Library	API integration & async data fetching	Used to fetch ML predictions from Flask backend into React frontend	Implemented efficient data flow and ensured real-time updates

Appendix 3.A: Installation Guide

This guide provides the necessary instructions to deploy the SafeSight Workplace Safety Management System in a local development environment. The project is divided into three primary tiers: the MySQL Database, the Node.js Backend, and the React Frontend, with a specialized Python AI Service.

A.1 Prerequisites

Ensure the following software is installed on the host machine:

- **Node.js** (v16.0 or higher) & **npm**
- **Python** (3.8 or higher)
- **MySQL Server** (v8.0 or higher)
- **Git**

A.2 Database Setup (MySQL)

1. Open your MySQL terminal or Workbench.
2. Create a new database:

SQL

```
CREATE DATABASE safesight_db;
```

3. Import the provided schema file (/BackEnd/config/schema.sql) to generate the required relational tables.
4. Update the .env file in the /BackEnd directory with your local credentials:

Code snippet

```
DB_HOST=localhost  
DB_USER=root  
DB_PASS=xxxxxxxxx  
DB_NAME=safesight_db
```

A.3 AI Service Configuration (Python)

The predictive engine requires a virtual environment to manage machine learning dependencies.

1. Navigate to the AI directory: `cd AI_Service`
2. Create a virtual environment:

```
python -m venv venv
```

3. Activate the environment:

- **Windows:** venv\Scripts\activate
- **Mac/Linux:** source venv/bin/activate

4. Install dependencies:

```
pip install -r requirements.txt
```

```
python predict.py
```

A.4 Backend and Frontend Deployment

SafeSight requires both the server and the client-side application to be running simultaneously.

Step 1: Start the Backend Server

1. Navigate to the backend directory: `cd BackEnd`
2. Install Node.js dependencies:

```
npm install
```

3. Start the server:

```
node server.js
```

The server will initialize on `http://localhost:5000`

Step 2: Start the Frontend Application

1. Navigate to the frontend directory: `cd FrontEnd`
2. Install React dependencies:

```
npm install
```

3. Launch the dashboard:

```
npm start
```

The application will open automatically in your browser at `http://localhost:3000`

A.5 Troubleshooting

- **CORS Errors:** Ensure the BackEnd/server.js file is configured to allow requests from localhost:3000.
- **Model Errors:** Ensure the predict.py script in the AI_Service has read access to the dataset path defined in the script.

- **Database Connection:** Verify that the MySQL service is running before starting the Node.js server.

Appendix 4.B: User Guide

(NOTE: The guide with screenshots available in the DOCUMENTS REPORT folder)

This guide is designed for the administrative and safety personnel at the Fairmont Waterfront. It provides a step-by-step walkthrough of the SafeSight Command Center, enabling users to manage real-time incidents and leverage AI-driven insights for long-term hazard prevention.

B.1 Accessing the Command Center

Upon launching the application, users are presented with the SafeSight Home page. From the navigation choose the role. In the Dashboard the navigation bar at the top allows users to toggle between the two primary operational modes:

1. **Operations:** For real-time monitoring and administrative action.
2. **AI Intelligence:** For strategic planning and predictive analysis.

B.2 Managing Real-Time Safety (Operations View)

The Operations Control screen is the primary interface for daily safety management.

1. **Monitor KPIs:** View the "Total Logs," "Critical Pending," and "Hospitalized" cards at the top for an immediate status update of the facility.
2. **Triage Incidents:** Use the Active Safety Observations table to identify urgent risks. Look for the Red "Critical" or Orange "High" labels.
3. **Apply AI Recommendations:** For each active incident, read the "AI Recommendation" column. These are prescribed immediate actions (e.g., *"Deploy Wet Floor signage"*) to mitigate further risk.
4. **Resolve Records:** Once an incident has been addressed on the floor and reviewed by HR, click the "Resolve" button to update the database and clear the item from the active dashboard.

5. **Search & Filter:** Use the search bar to find specific departments (e.g., "Kitchen") or filter the view by severity to focus on high-priority tasks.

B.3 Strategic Planning (AI Intelligence View)

The AI Strategic Intelligence screen is used by safety directors to identify underlying trends.

1. **Analyze Risk Velocity:** Review the historical trend chart to identify months with high incident spikes. Use this data to increase staffing or safety audits during those periods.
2. **Departmental Profiling:** Inspect the Department Risk Profile bar chart. Focus on departments at the top of the list (e.g., Maintenance or Housekeeping) as they represent the highest risk zones.
3. **Utilize the Safety Radar:** Use the radar chart to visualize the "safety footprint" across all departments to ensure balanced resource allocation.

B.4 Proactive Risk Mitigation (Predictive Risk Engine)

This standalone module forecasts future hazards before they occur.

1. **Review 30-Day Predictions:** Each departmental card shows a Risk Probability bar. If the bar is in the "High" zone, immediate preventative measures are required.
2. **Execute Prescriptive Actions:** Read the "Strategic Action" section on the card. Unlike the operational view (which is reactive), these are proactive steps (e.g., *"Rotate heavy-duty tasks"*) intended to prevent predicted ergonomic strains or equipment failures.
3. **Confidence Check:** Note the "Prediction Confidence" percentage. Higher percentages (95%+) indicate a very high statistical likelihood based on the Random Forest model analysis.

B.5 Compliance and Exporting

To fulfill regulatory requirements for WorkSafeBC:

1. **Navigate to the Export Section:** Access the compliance log via the "WorkSafeBC Export" tab.

2. **Generate Individual Reports:** Click "Generate PDF" next to any resolved incident to create a pre-formatted report ready for submission.
3. **Perform Bulk Audits:** Use the "Export All to Excel" button to download the entire safety history for internal performance reviews or annual audits

B.6 User Roles and Responsibilities

SafeSight utilizes a role-based workflow to ensure a clear chain of command and data integrity.

B.6.1 The Department Manager (Front-Line Reporting)

The Department Manager is responsible for the immediate documentation of hazards and the execution of on-site safety protocols.

- **Primary Task:** Incident Reporting: Access the "Report Incident" form to document any workplace hazard or injury immediately.
- **Workflow:**
 1. Select the Location and Incident Type (e.g., Near Miss, Injury).
 2. Provide the Narrative in the "Sequence of Events" field. *Note: Be as descriptive as possible, as this text is analyzed by the AI to generate future recommendations.*
 3. Enter Witness Statements if applicable.
 4. Submit the form to sync the data with the MySQL backend.
- **Operational Duty:** Monitor the Operations View for their specific department and implement the AI Recommendations provided by the system.

B.6.2 HR & Safety Director (Audit and Resolution)

The HR and Safety Director role is responsible for the administrative lifecycle and high-level strategic oversight.

- **Primary Task:** Record Resolution: Review all incidents marked as *"Awaiting HR Review."* Once the incident has been investigated and the root cause analyzed, they click the "Resolve" button.

- **Strategic Task:** Resource Allocation: Monitor the AI Strategic Intelligence dashboard to identify which Department Managers require additional safety training or equipment based on high Risk Velocity or Severity Mix.
- **Compliance Duty:** Responsible for using the WorkSafeBC Export module to generate the official documentation required for external regulatory bodies.

B.7 Summary of System Workflow

1. **Incident Occurs:** Department Manager fills out the Report Incident form.
2. **Data Processing:** The Node.js Backend saves the report to and analyzes the record to update the Predictive Risk Engine.
3. **Real-Time Response:** The incident appears in the Operations View with a specific AI Recommendation.
4. **Strategic Review:** HR reviews the incident and the AI Intelligence charts to see if this is a recurring trend.
5. **Closure:** HR clicks "Resolve" and generates a WorkSafeBC PDF for the legal file.